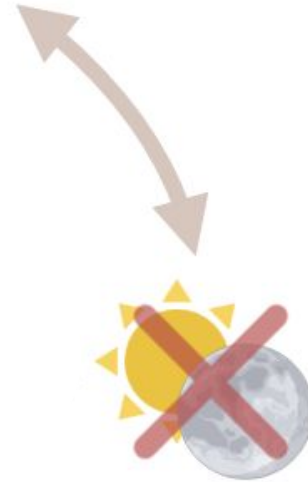


The effects of stress on gene expression in adipose tissues

Yana AlHebri
Dipesh Chaudhury

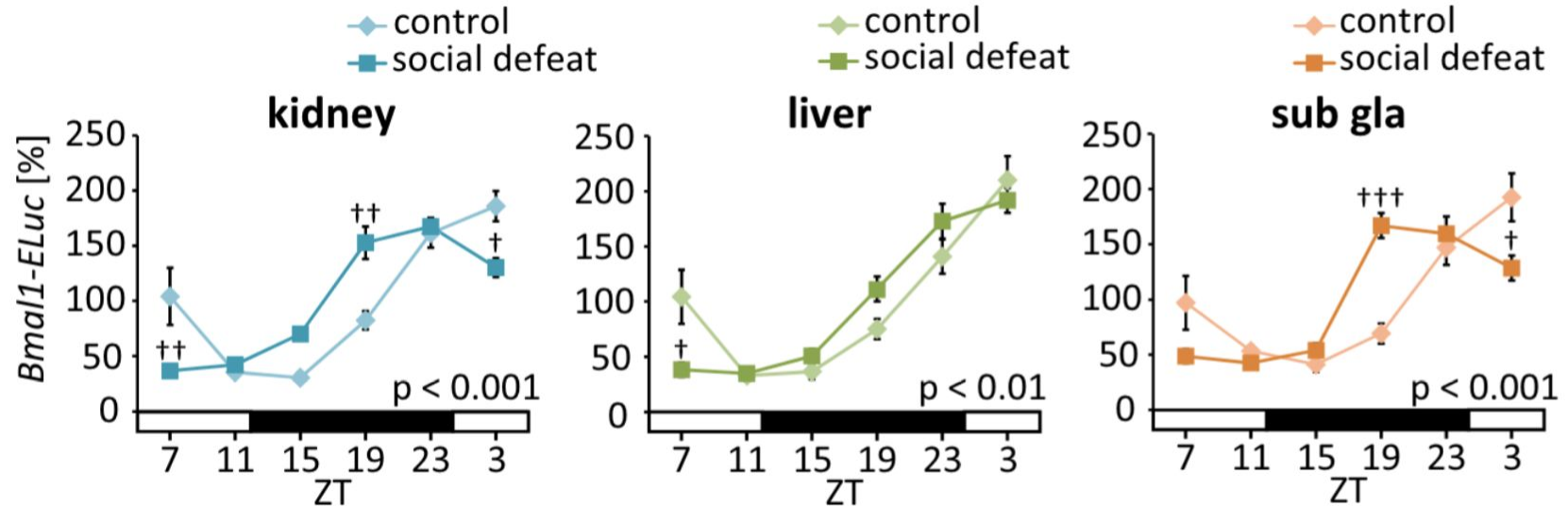
Chronic stress, circadian rhythms, and metabolism are connected



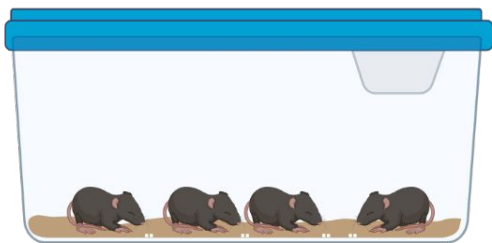
Van Der Kooji et al., *Mol. Cell. Neurosci.*, 2020

Fory et al., *Endocr. Rev.*, 2018

Stress causes phase shifts of the clock gene *Bmal1*'s expression in peripheral tissues

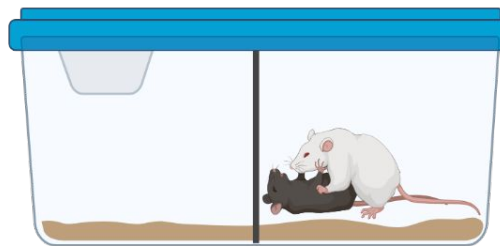


There are three main types of adipose tissue

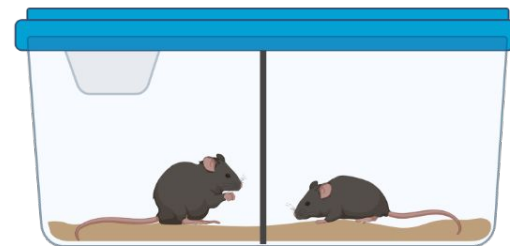


Homecage control (HC)

The mice are undisturbed

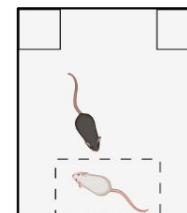
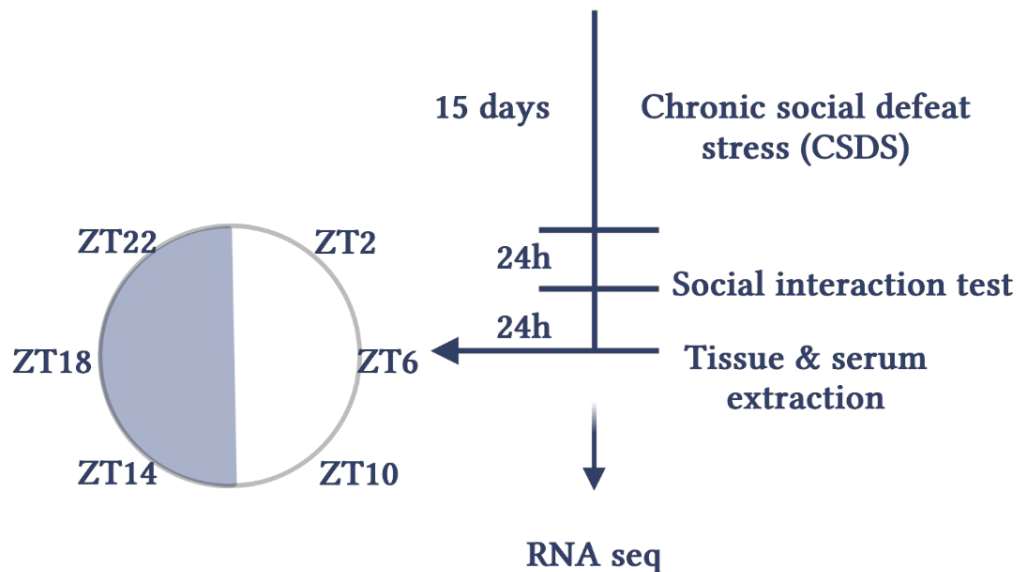


Chronic Social Defeat Stress (CSDS)



Chronic Mild Variable Stress (CMVS)

The mice experiences the same housing and daily handling conditions as CSDS mice.



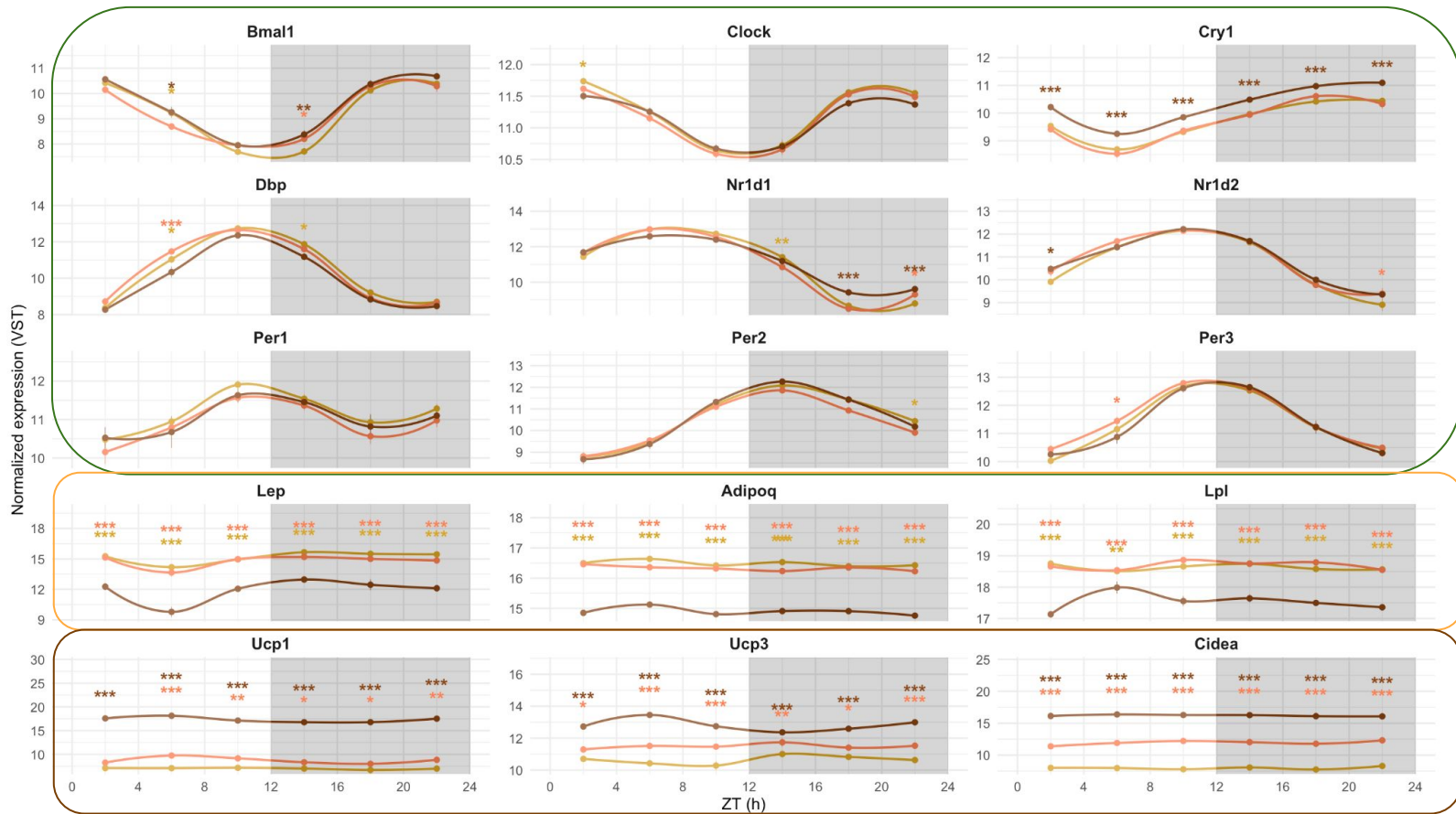
Question 1:

What is the baseline rhythmic gene expression in adipose tissues?

Data was validated by confirming known expression patterns in Control

Core Clock & Adipose Markers — Control, multi-tissue

Tissue vW scW BAT

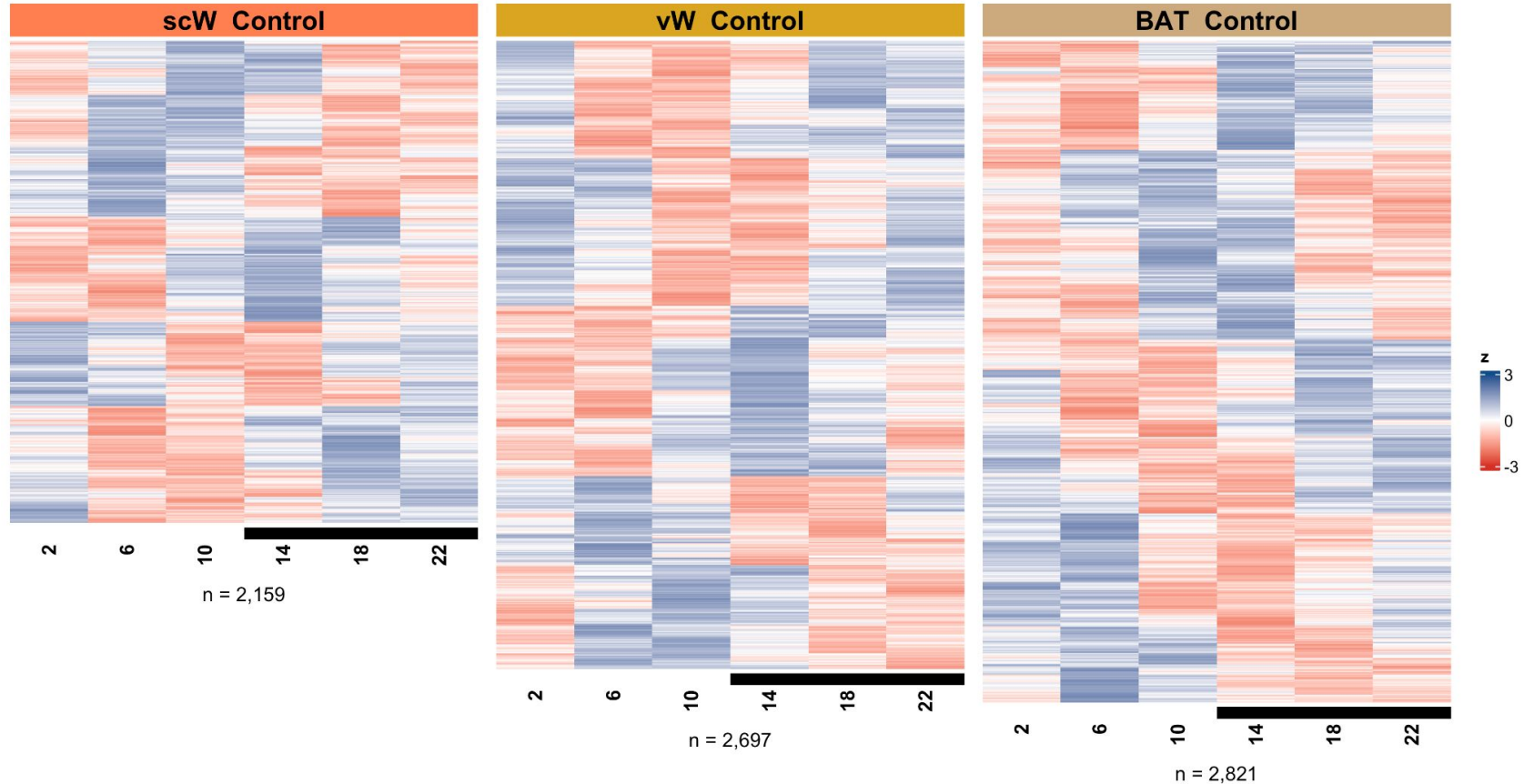


Circadian

Metabolism

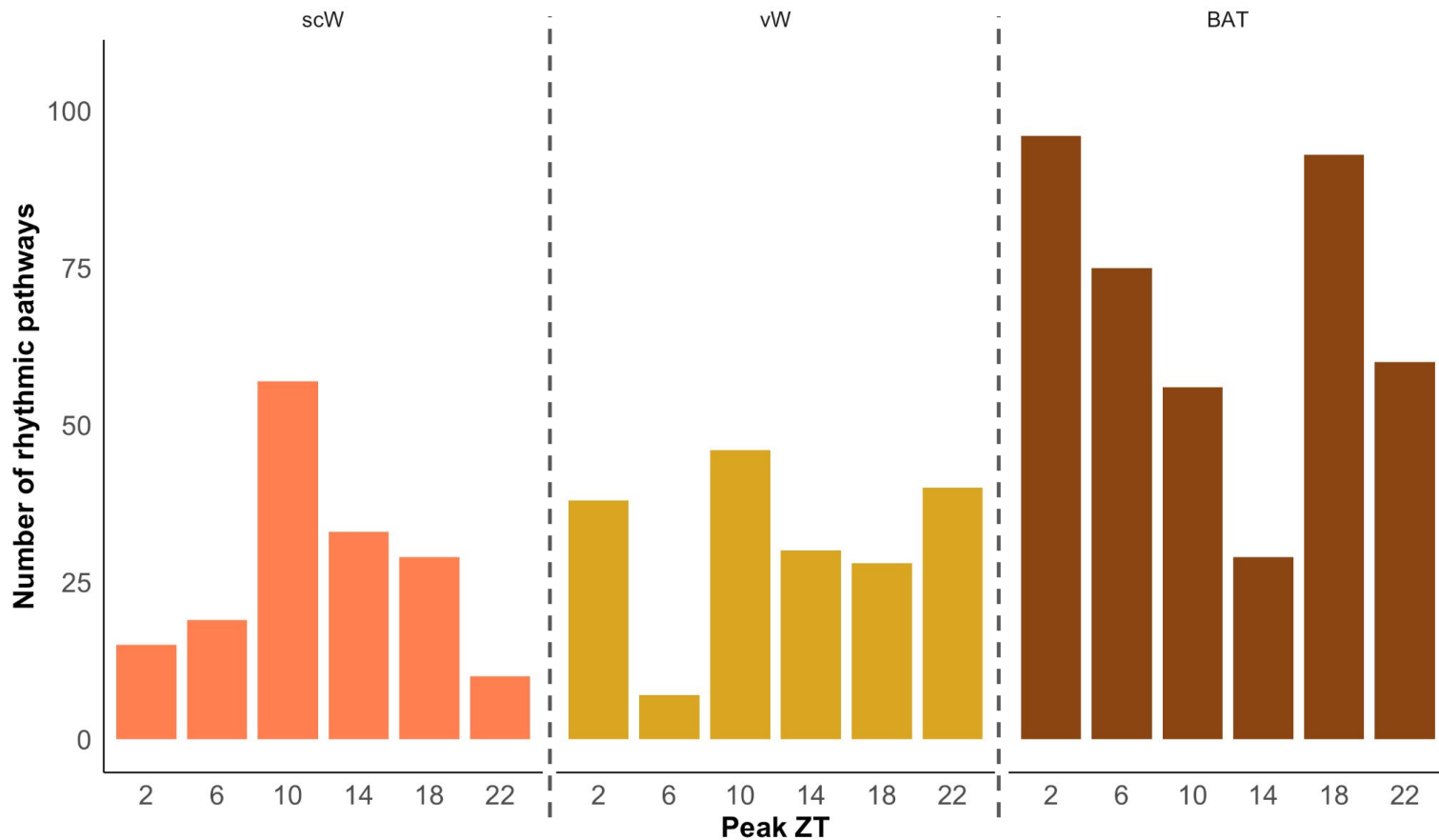
Thermogenesis

11% of sequenced genes are rhythmic in scW, 14% in vW, and 16.4% in BAT



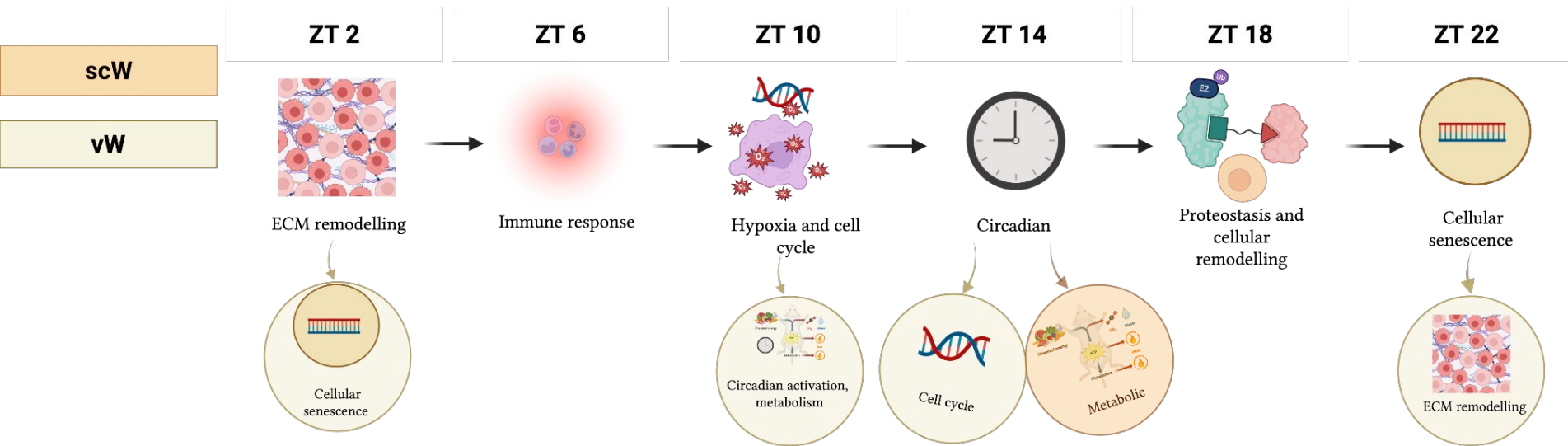
scW and BAT have opposite patterns of pathway peaks

Control Rhythmic Reactome pathway peaks by ZT



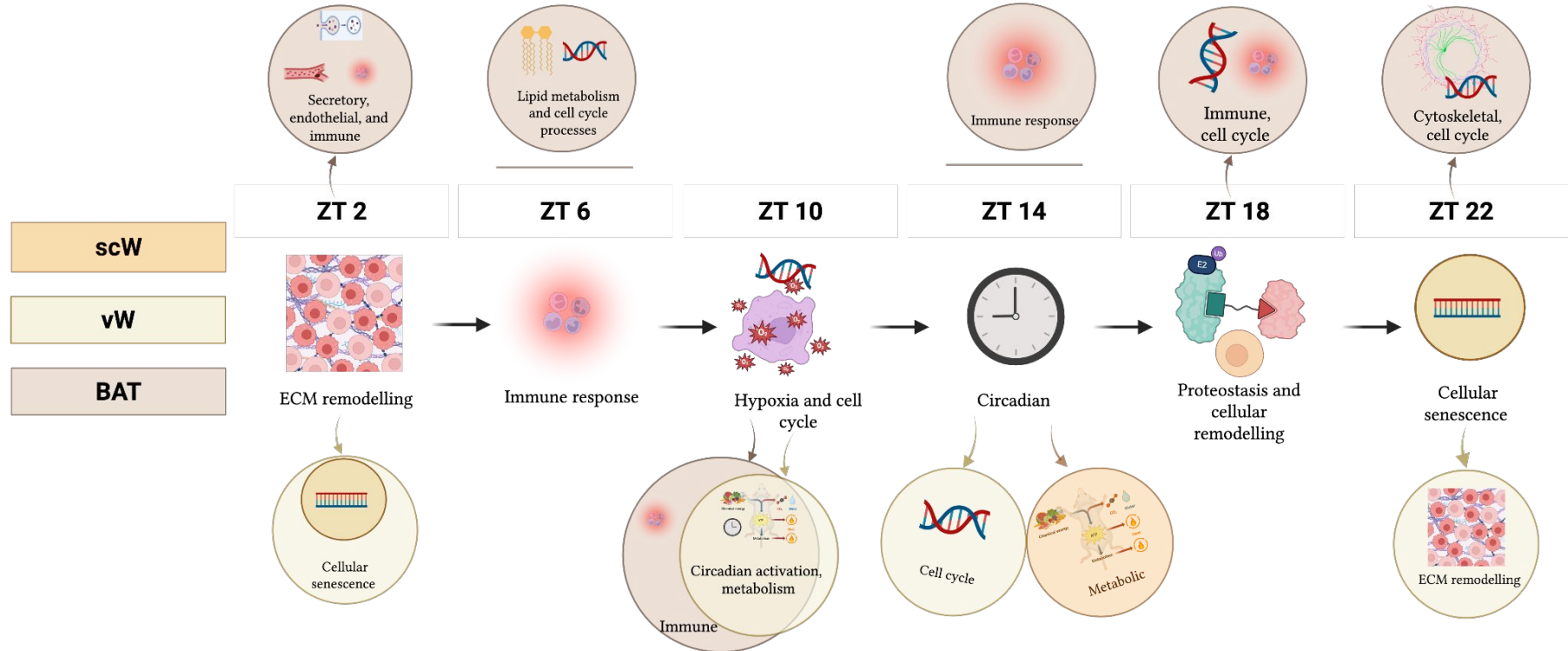
White adipose tissues have a similar pattern of rhythmic pathways

Reactome rhythmic pathways peaking in Control group at each timepoint



The BAT has more rhythmic pathways

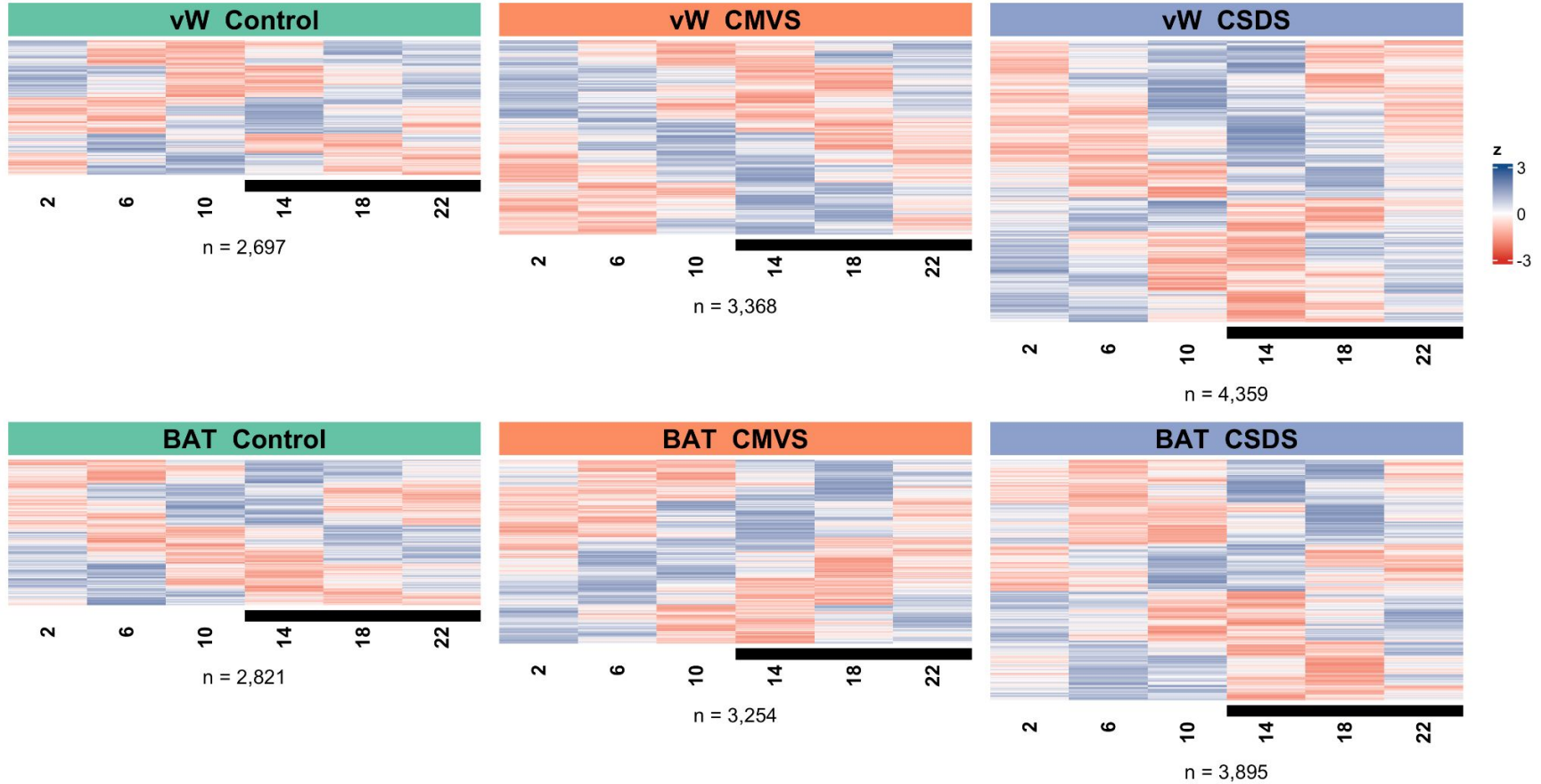
Reactome rhythmic pathways peaking in Control group at each timepoint



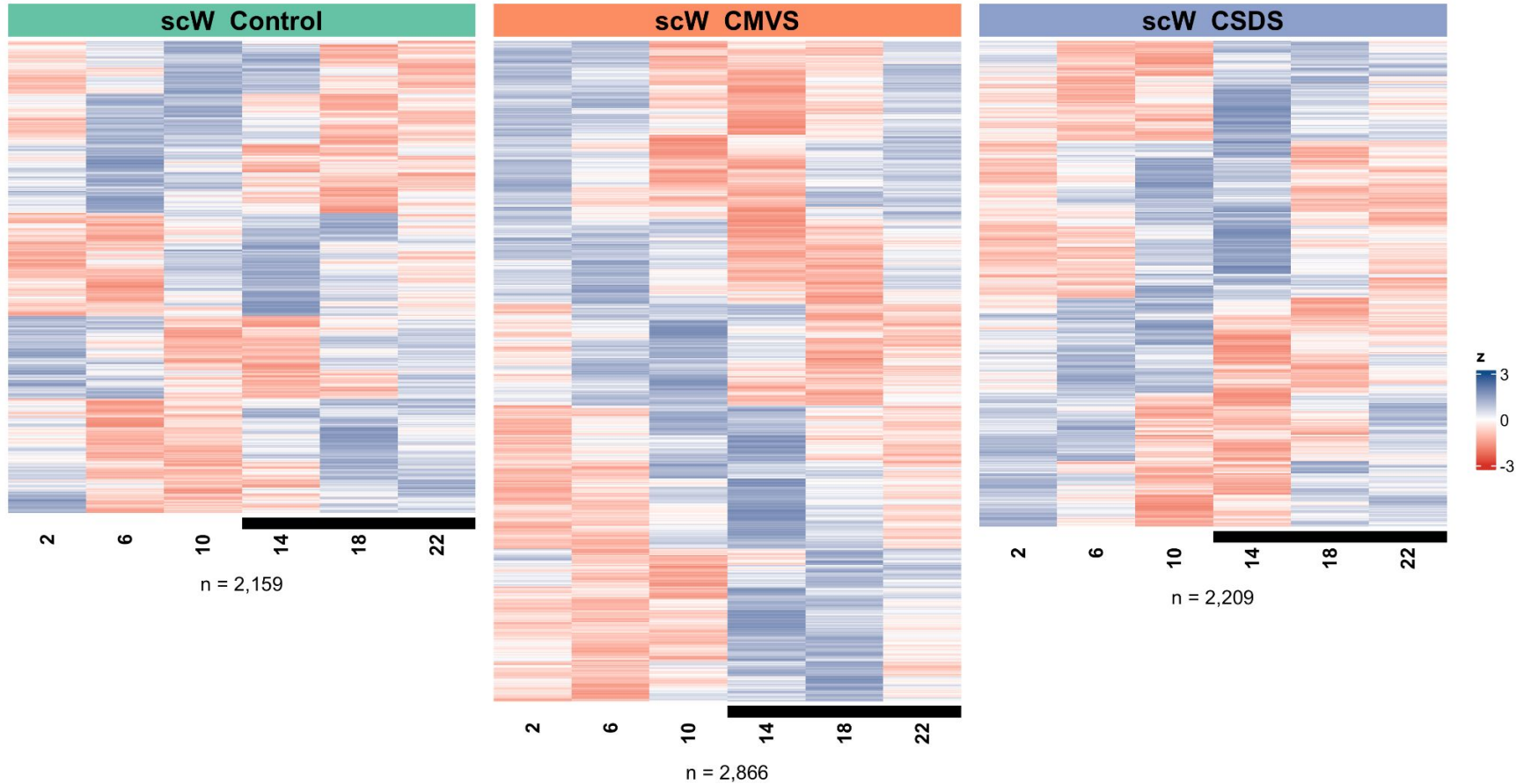
Question 2:

What are the effects of weak and strong stress on the rhythmic profiles of adipose tissues?

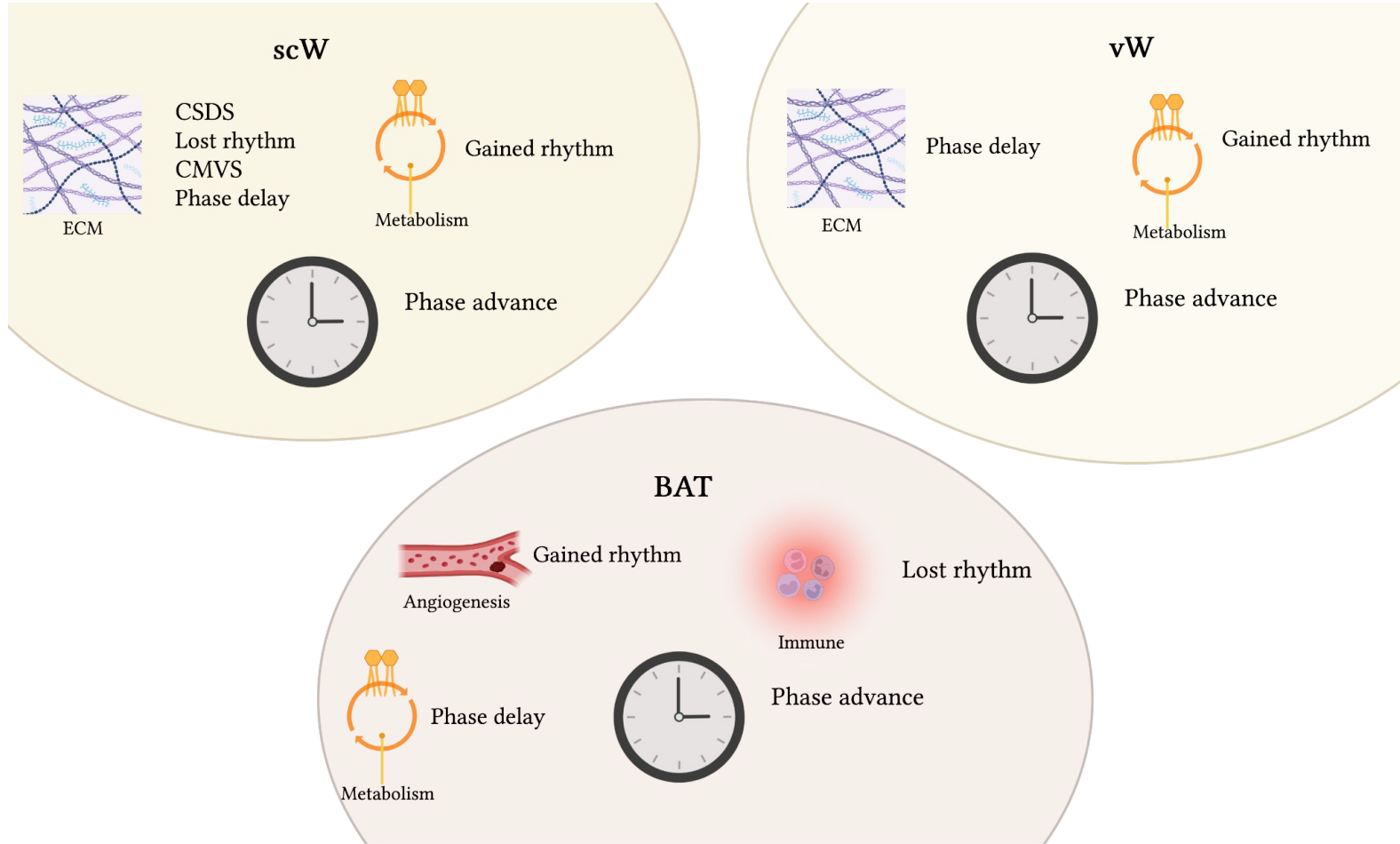
CSDS showed the highest number of rhythmic genes in vW and BAT



CMVS had the highest number of rhythmic genes in scW



Stress affected rhythmicity in all ATs, especially BAT



Question 3:

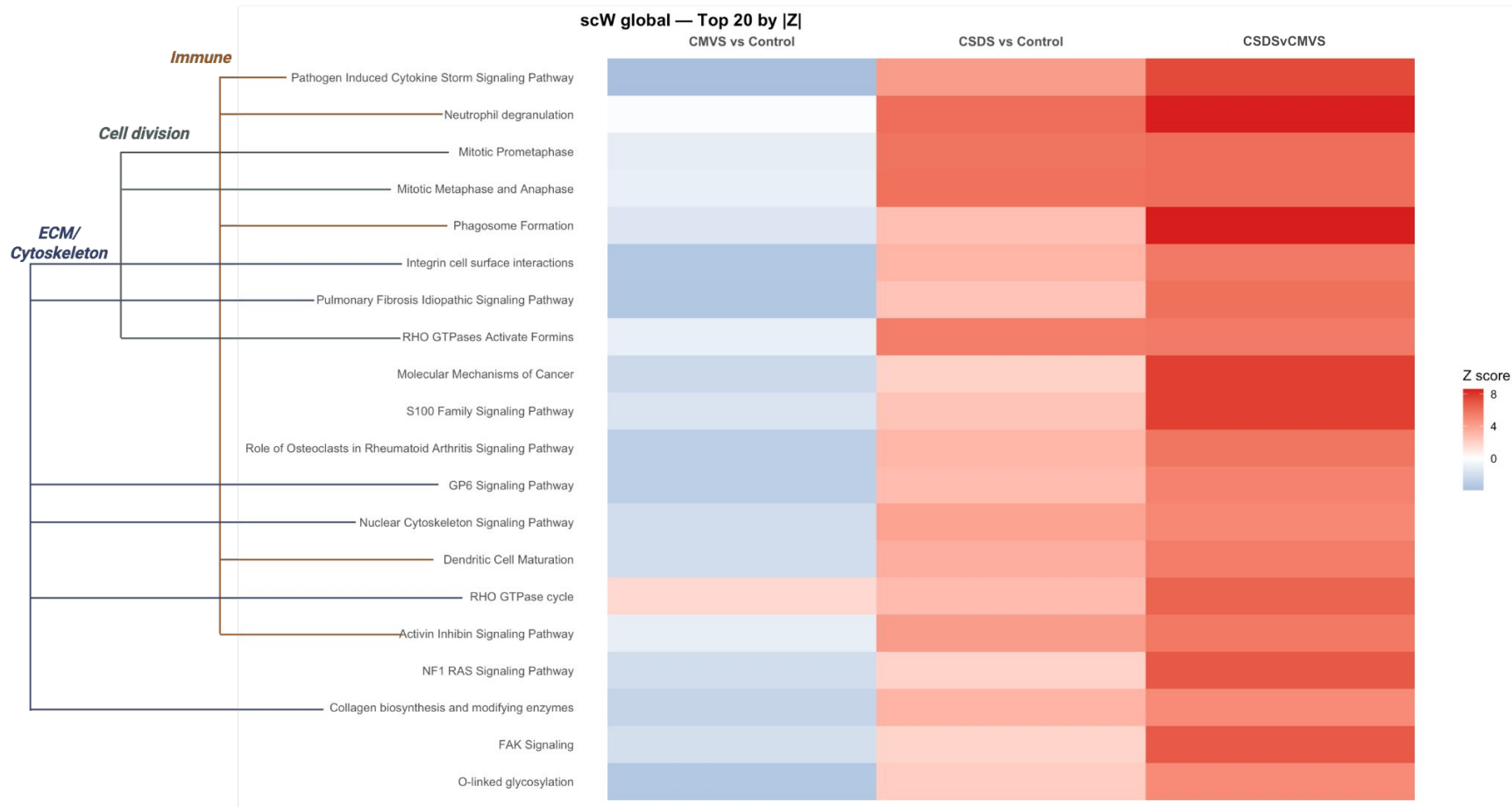
What are the differentially expressed genes in adipose tissues after weak and strong stress exposure?

The strong and weak stressors had different effects in scW

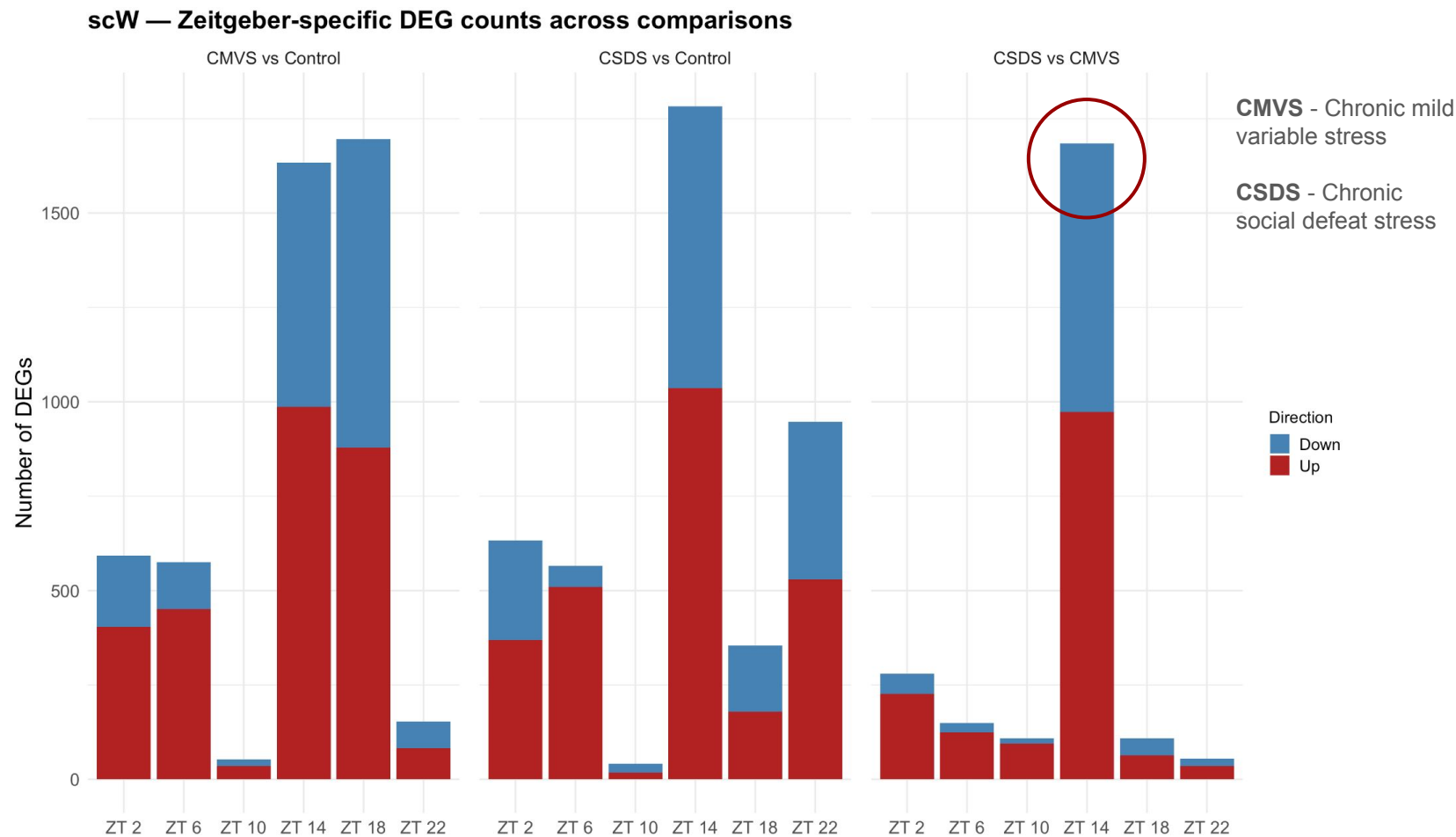
scW — Global/Day/Night DEG counts across comparisons



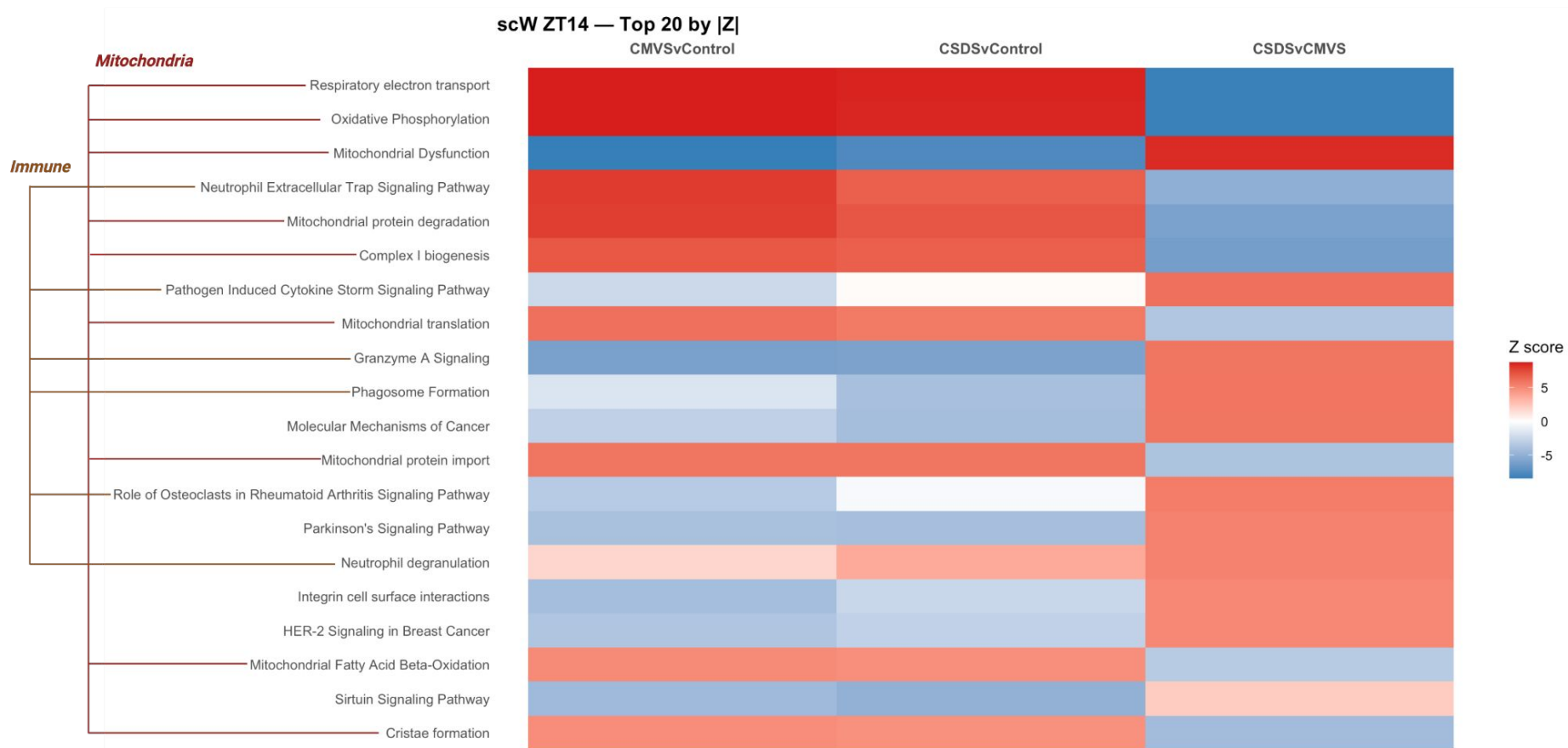
CSDS caused an ECM, immune, and cell division activation in scW



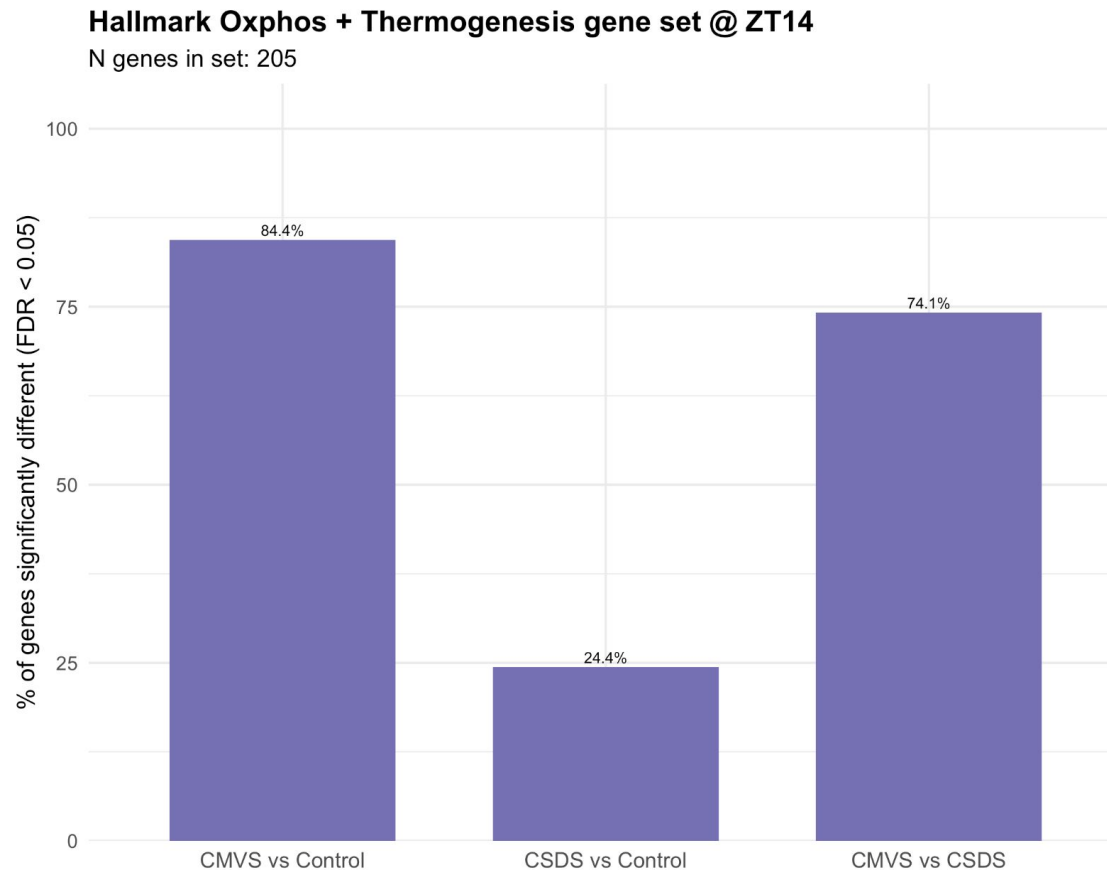
CSDS exhibited the strongest effect at zt14



CMVS caused mitochondrial activation at ZT14

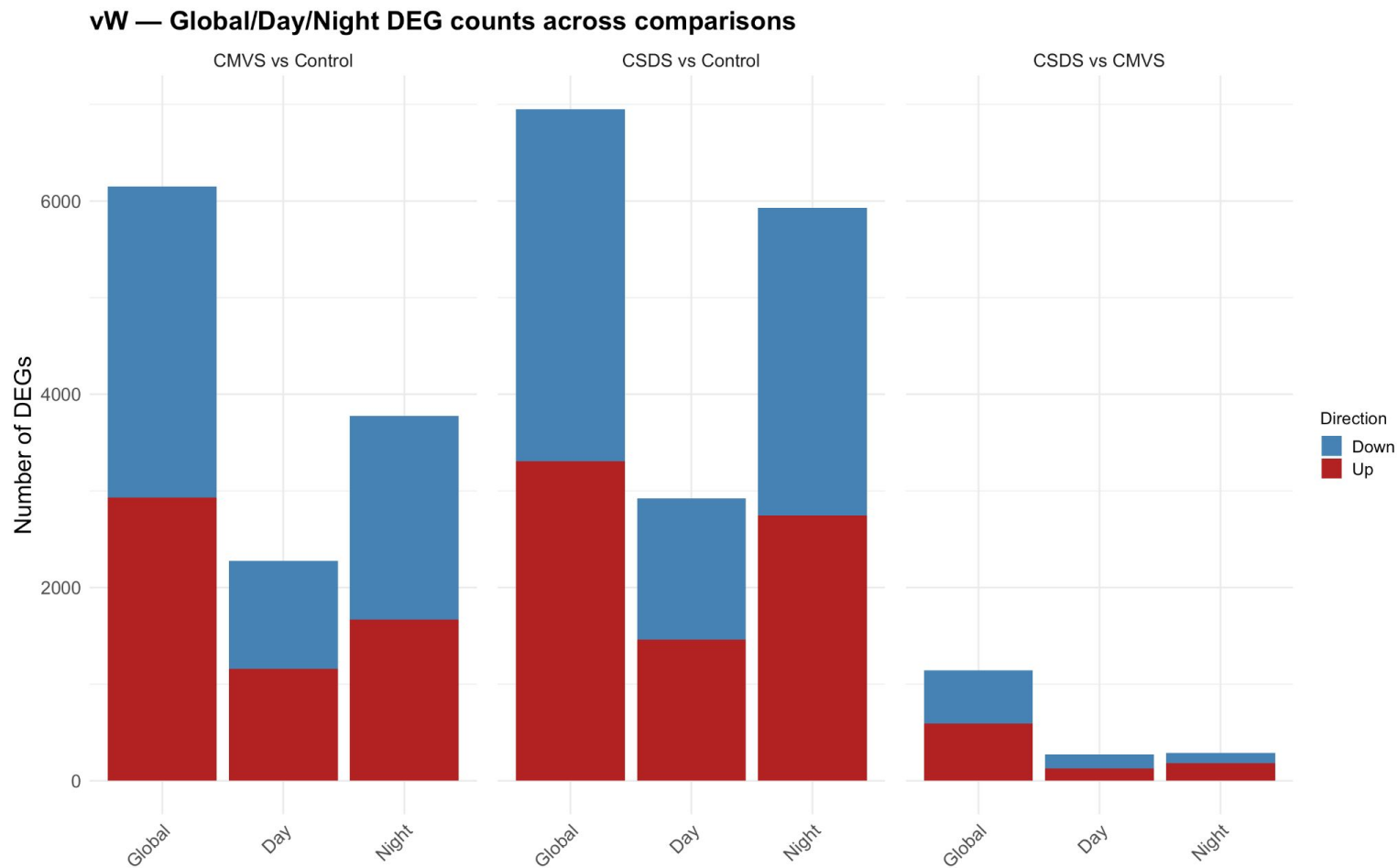


CMVS caused mitochondrial activation at ZT14

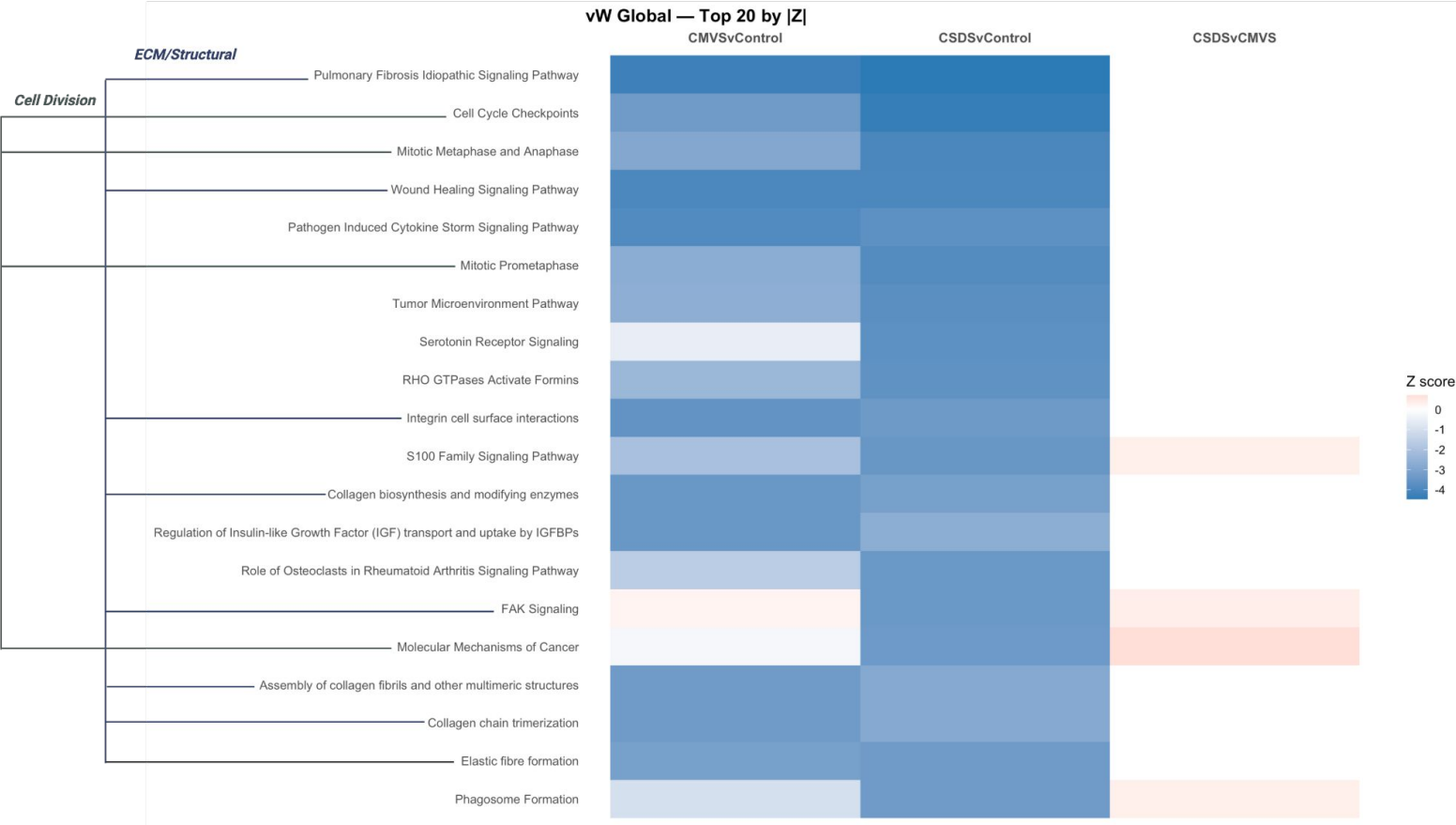


Visceral adipose tissue (vW)

CSDS and CMVS had mild differences in DEG count

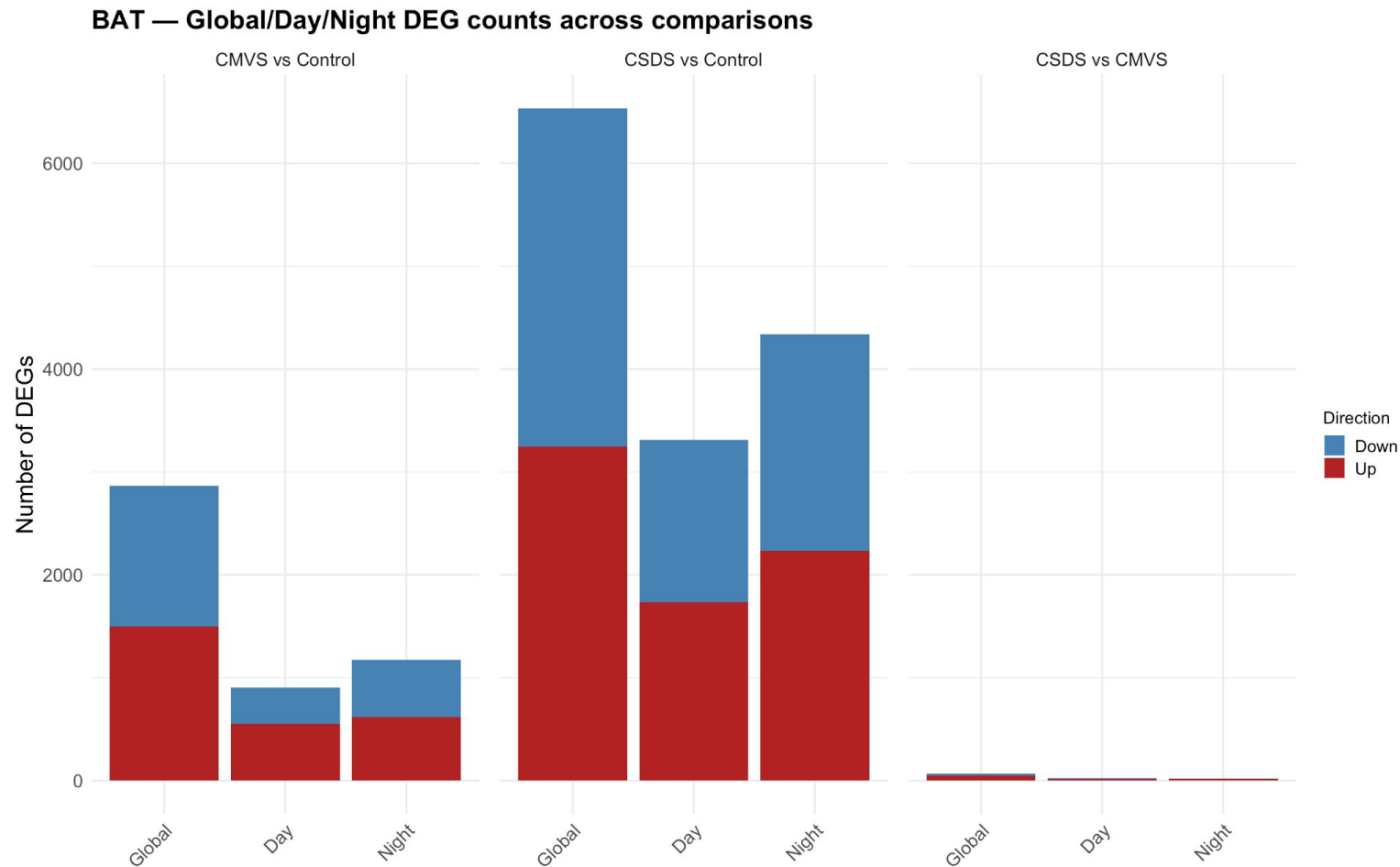


Stress caused reduced activation of ECM and Cell division pathways

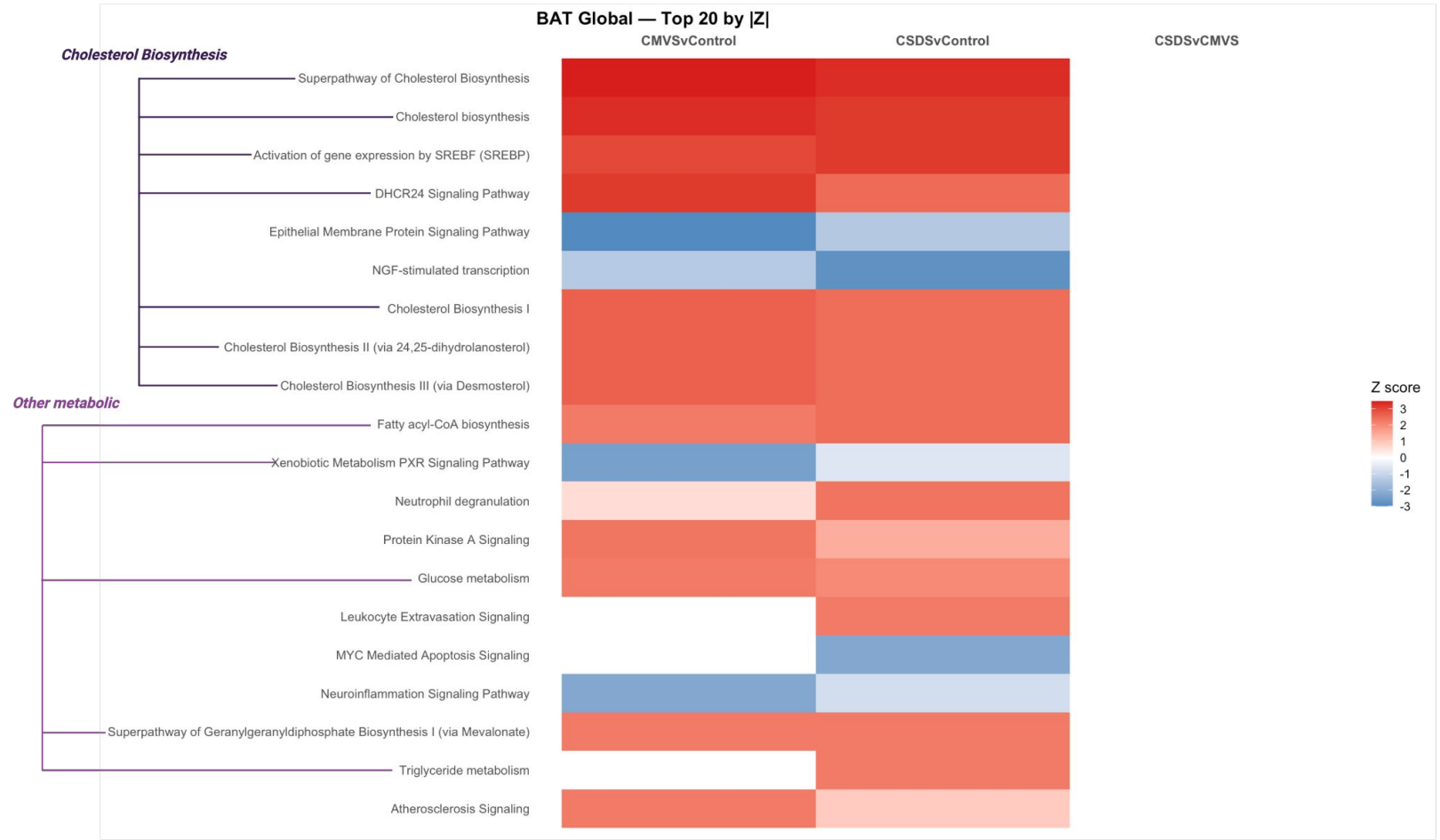


Brown adipose tissue (BAT)

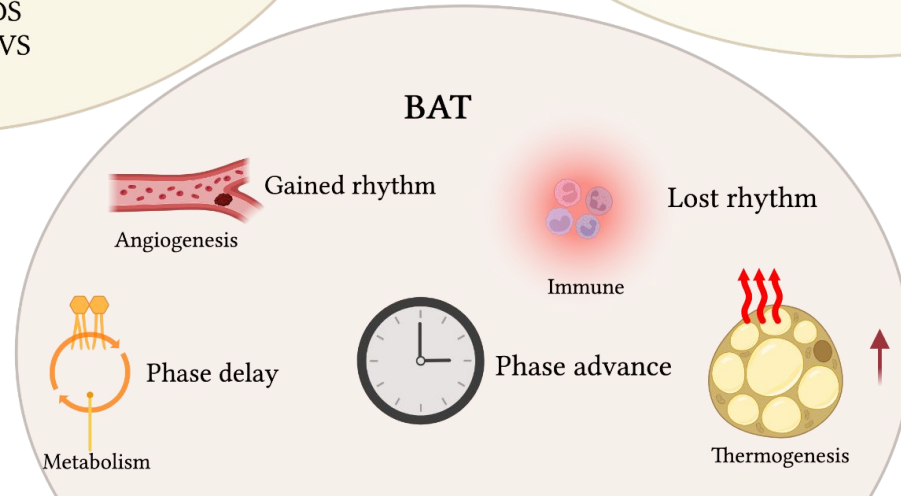
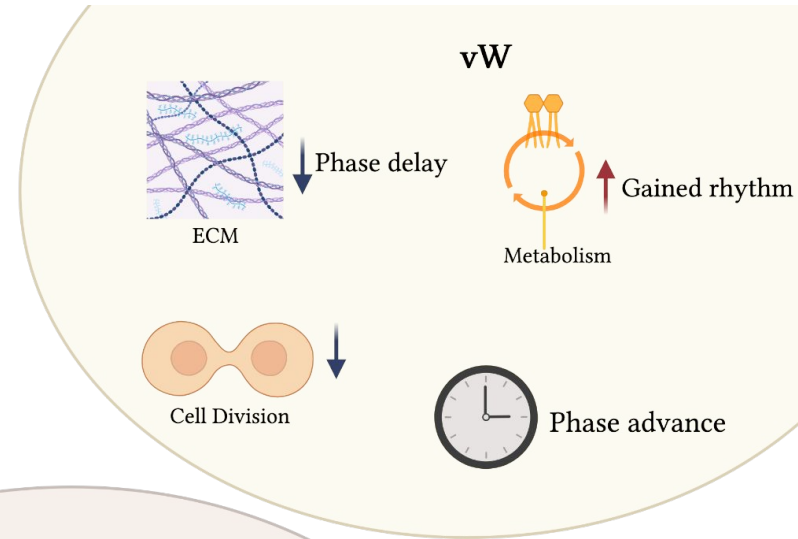
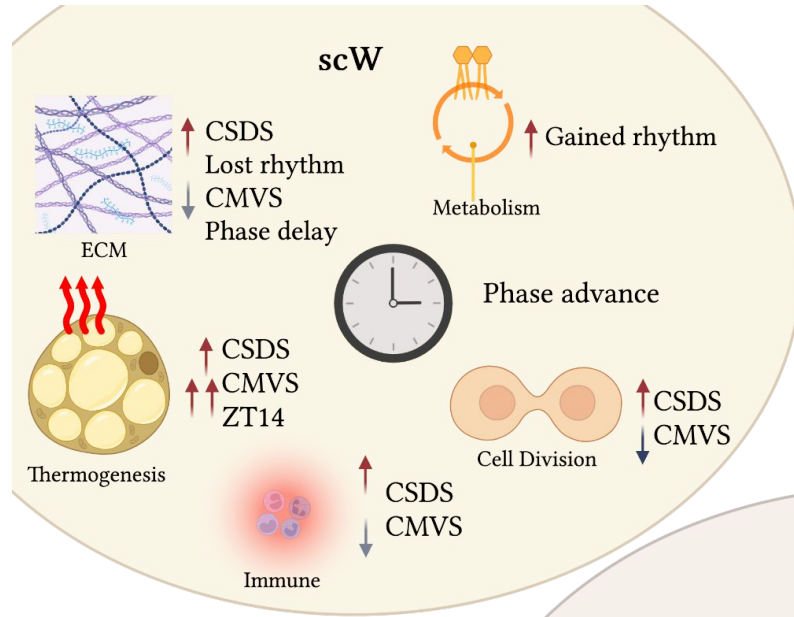
Low number of DEG between the strong and mild stress globally



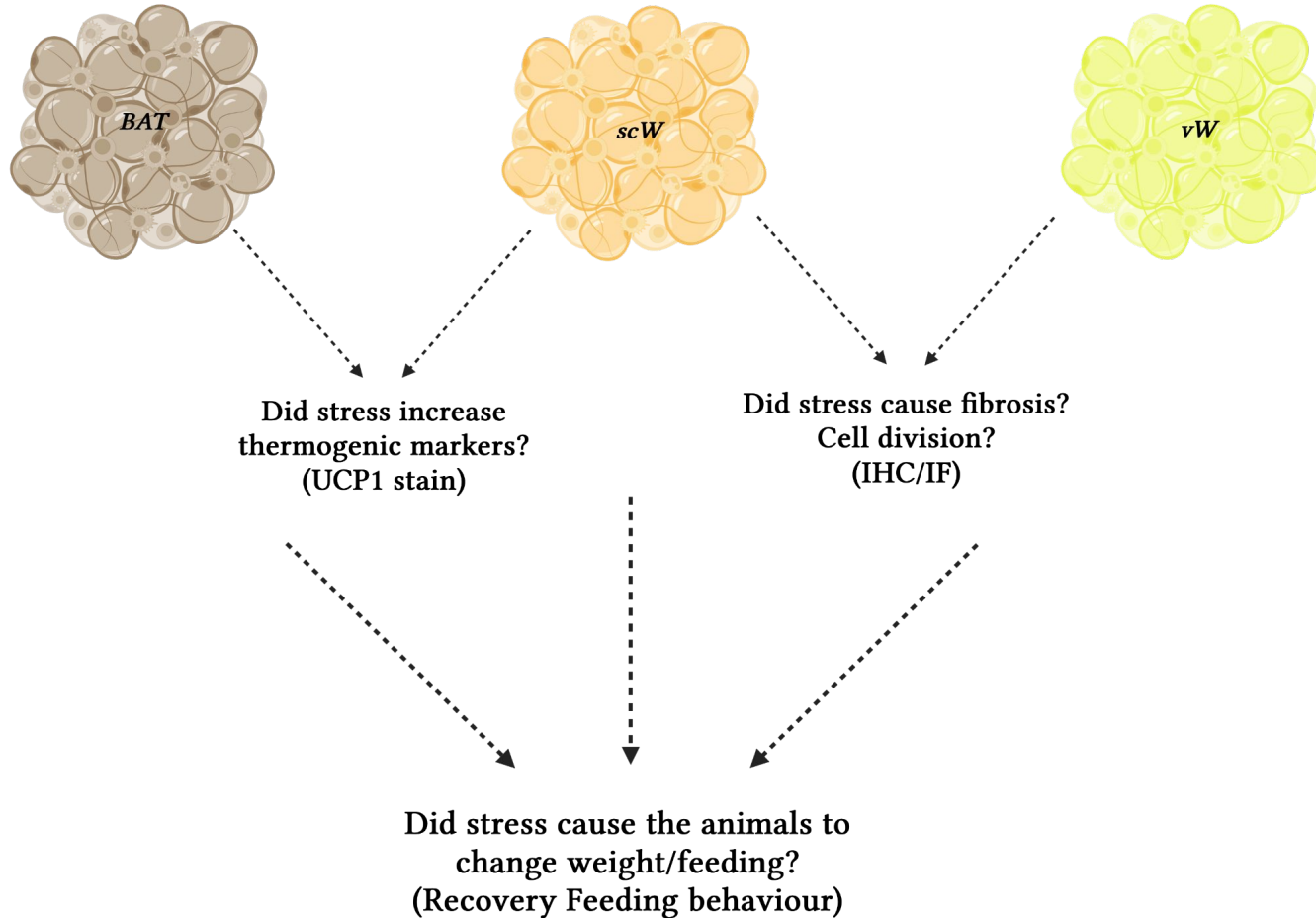
Both stresses caused activation in metabolic pathways



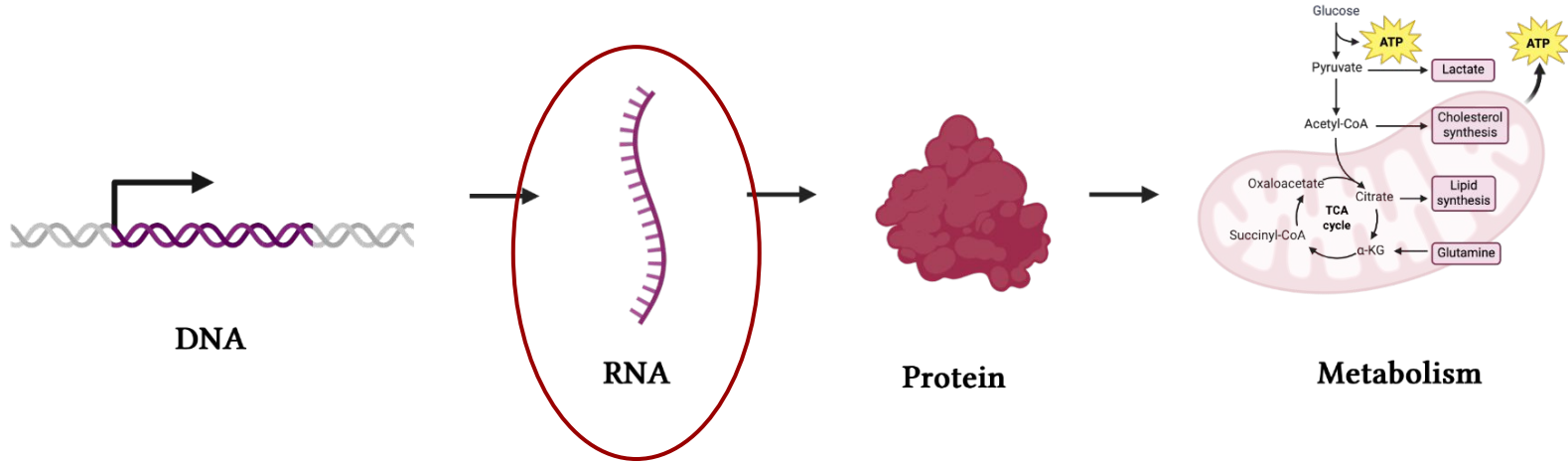
scW experienced stress-specific effects



Validation experiments

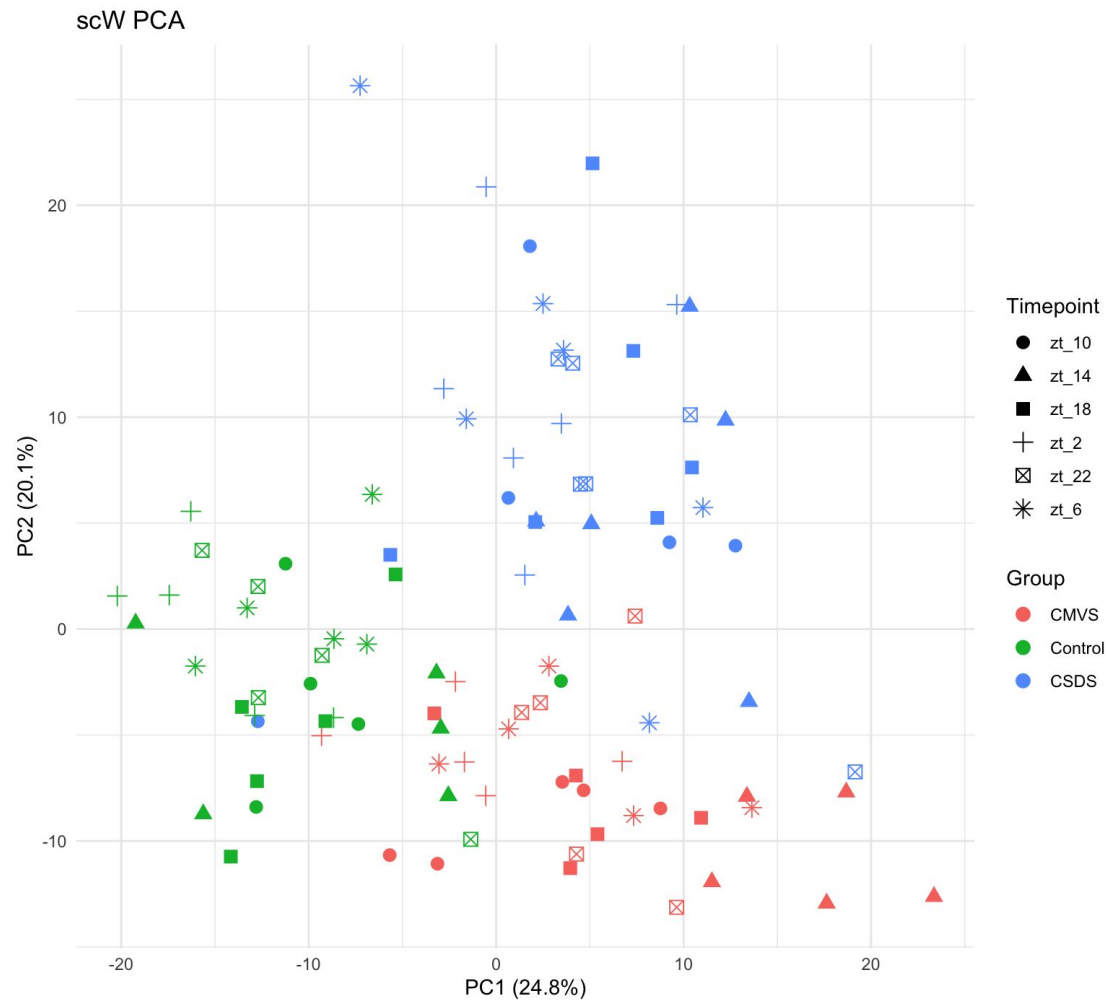


Follow up experiments

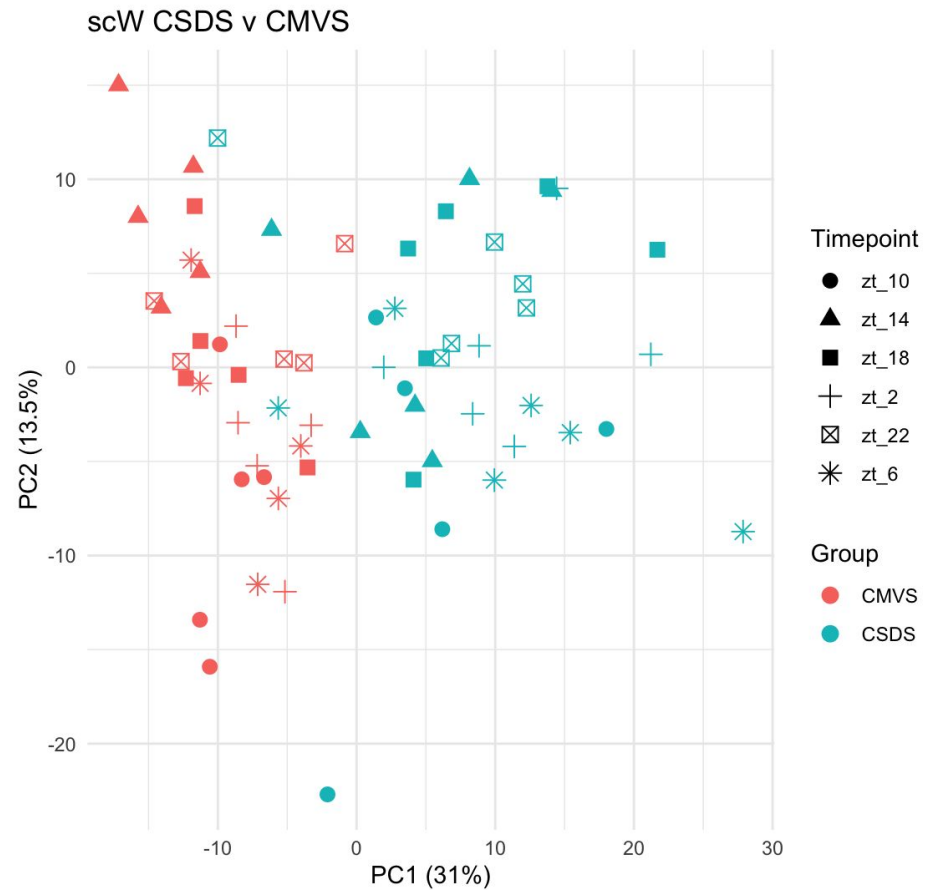
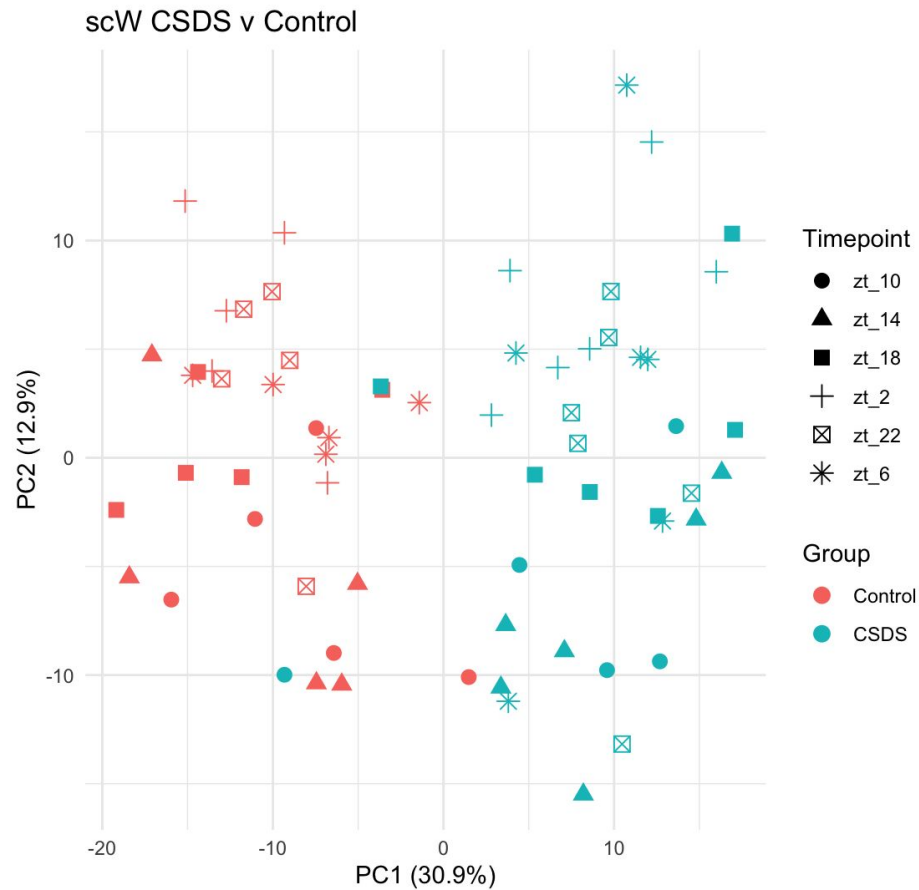


Thank you!

scW PCA

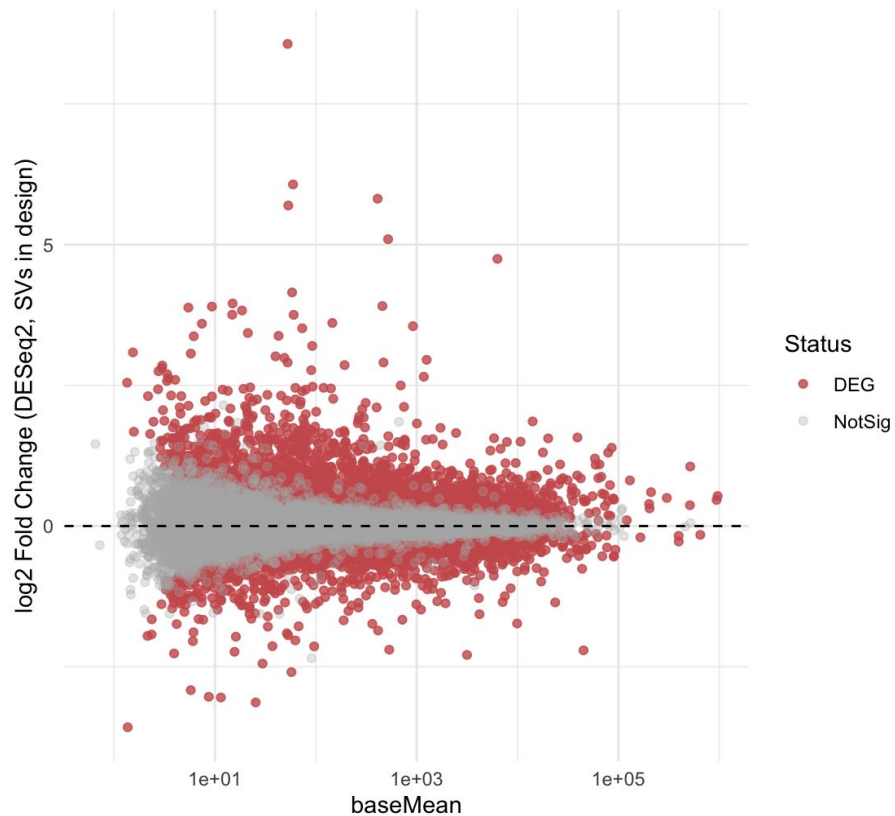


scW PCA pairwise

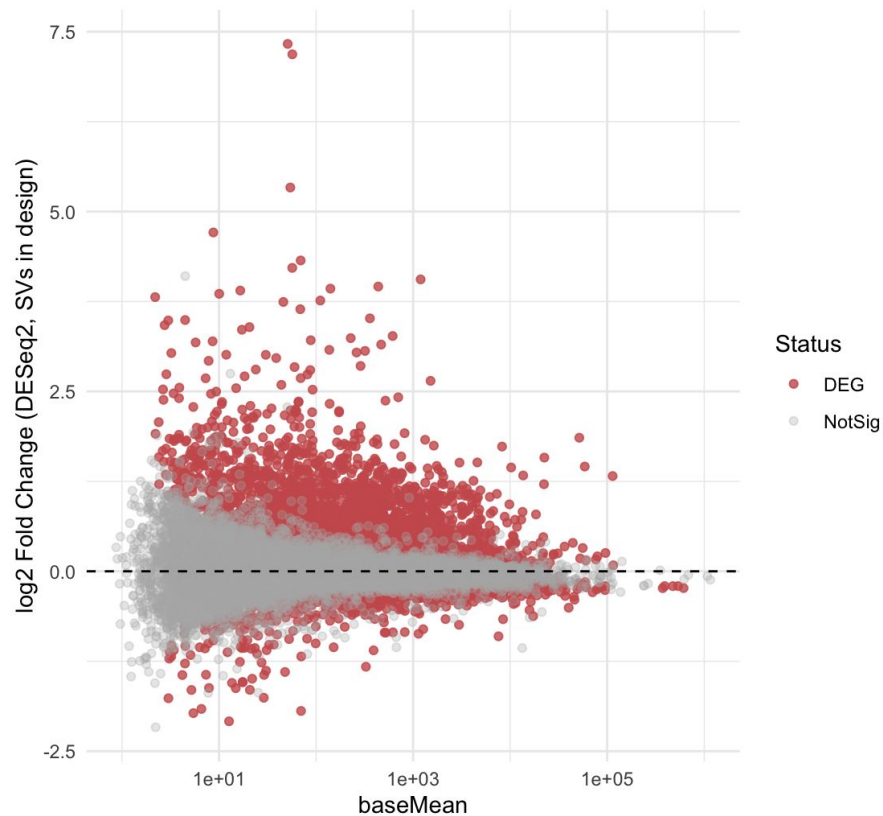


scW MA plots

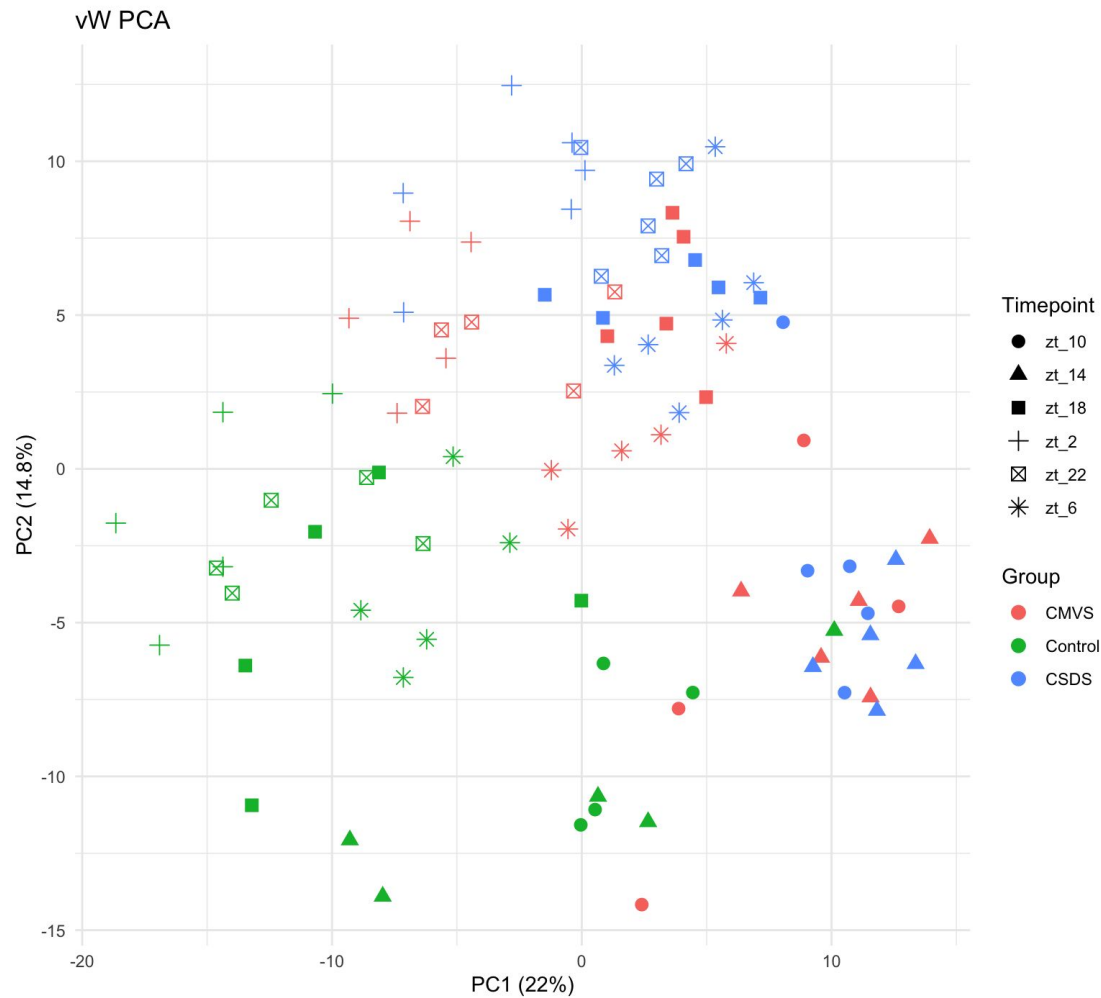
scW CSDS v Control



scW CSDS v CMVS

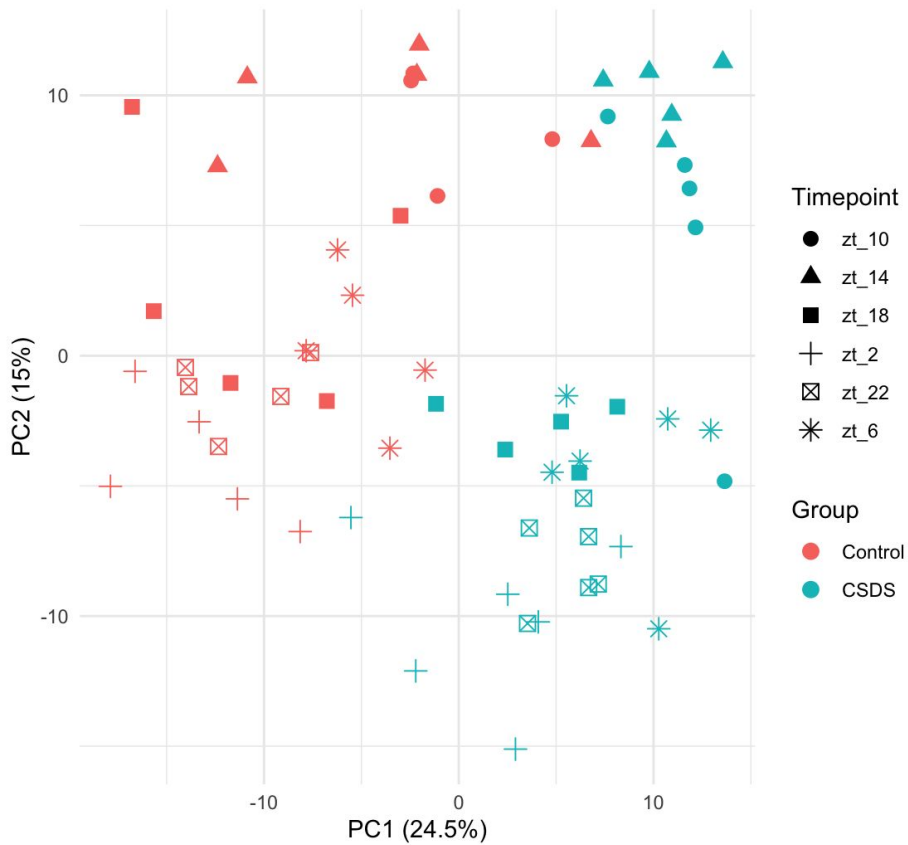


vW PCA

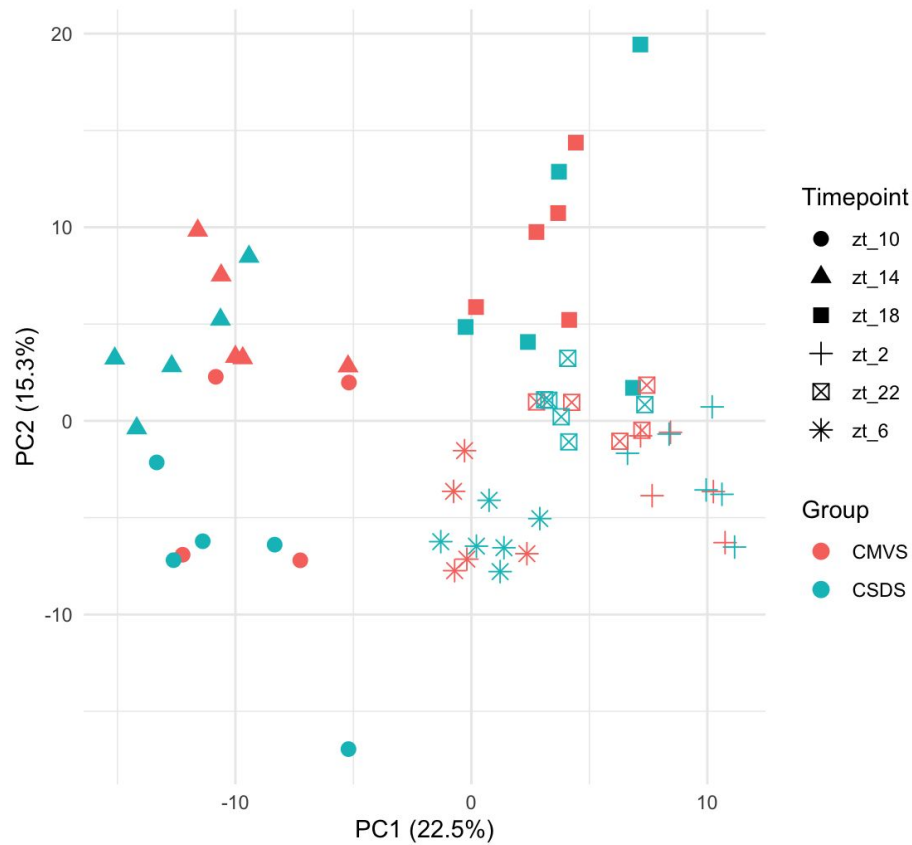


vW PCA pairwise

vW CSDS v Control

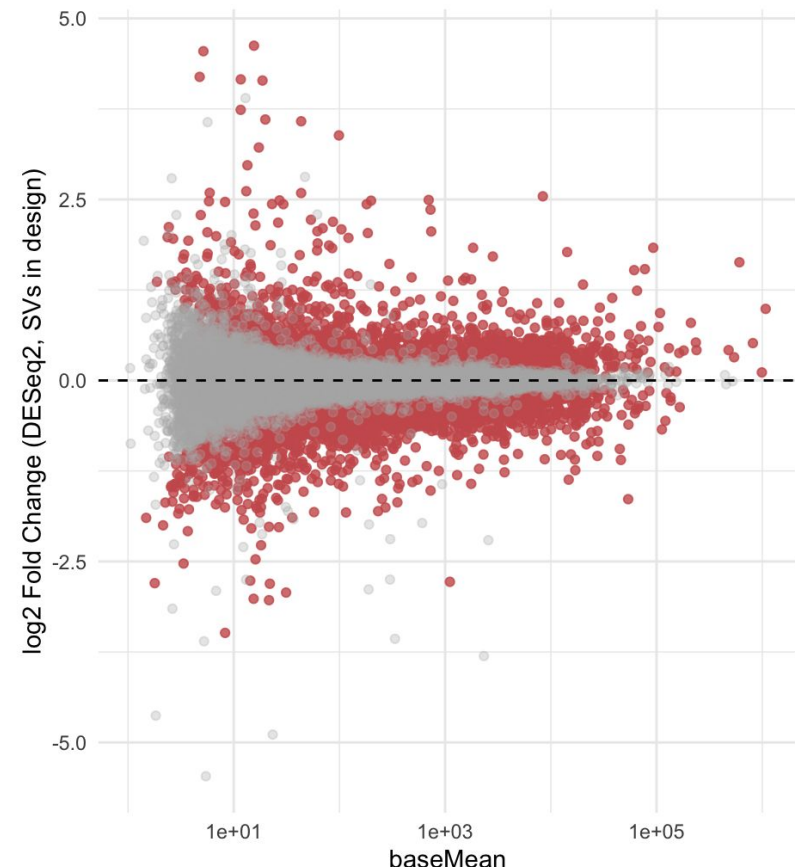


vW CSDS v CMVS

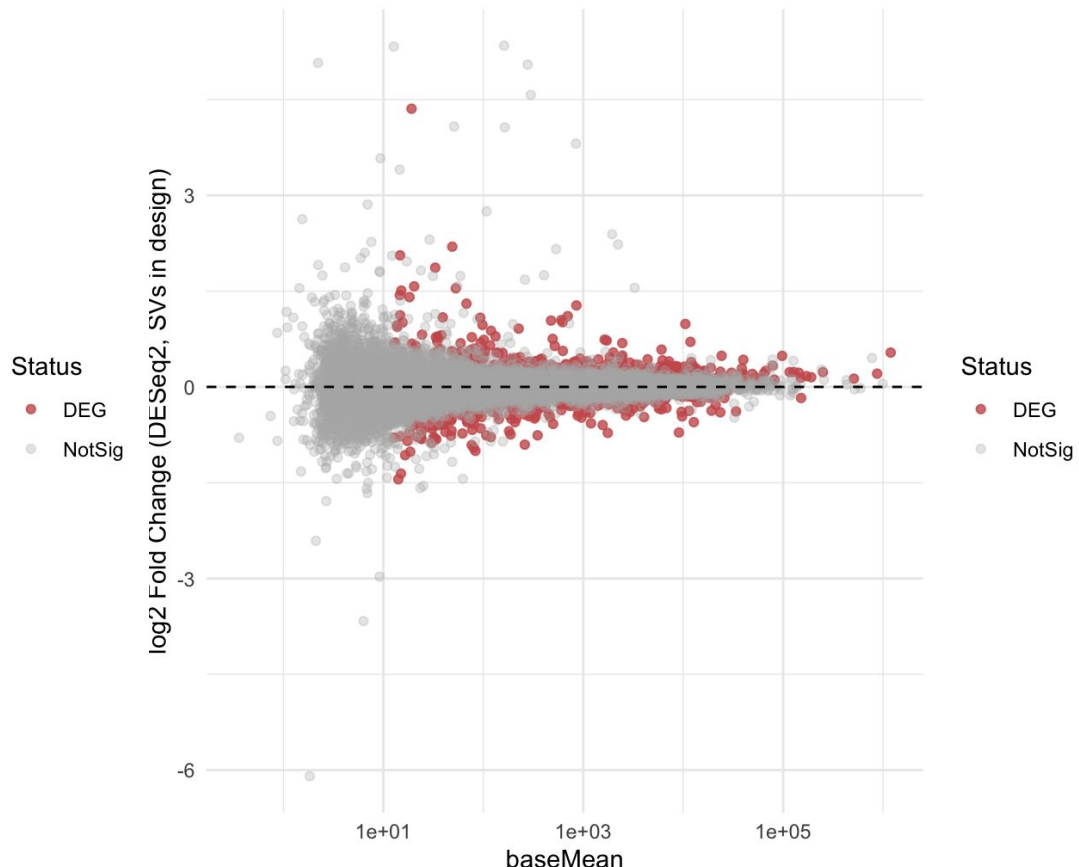


vW MA plots

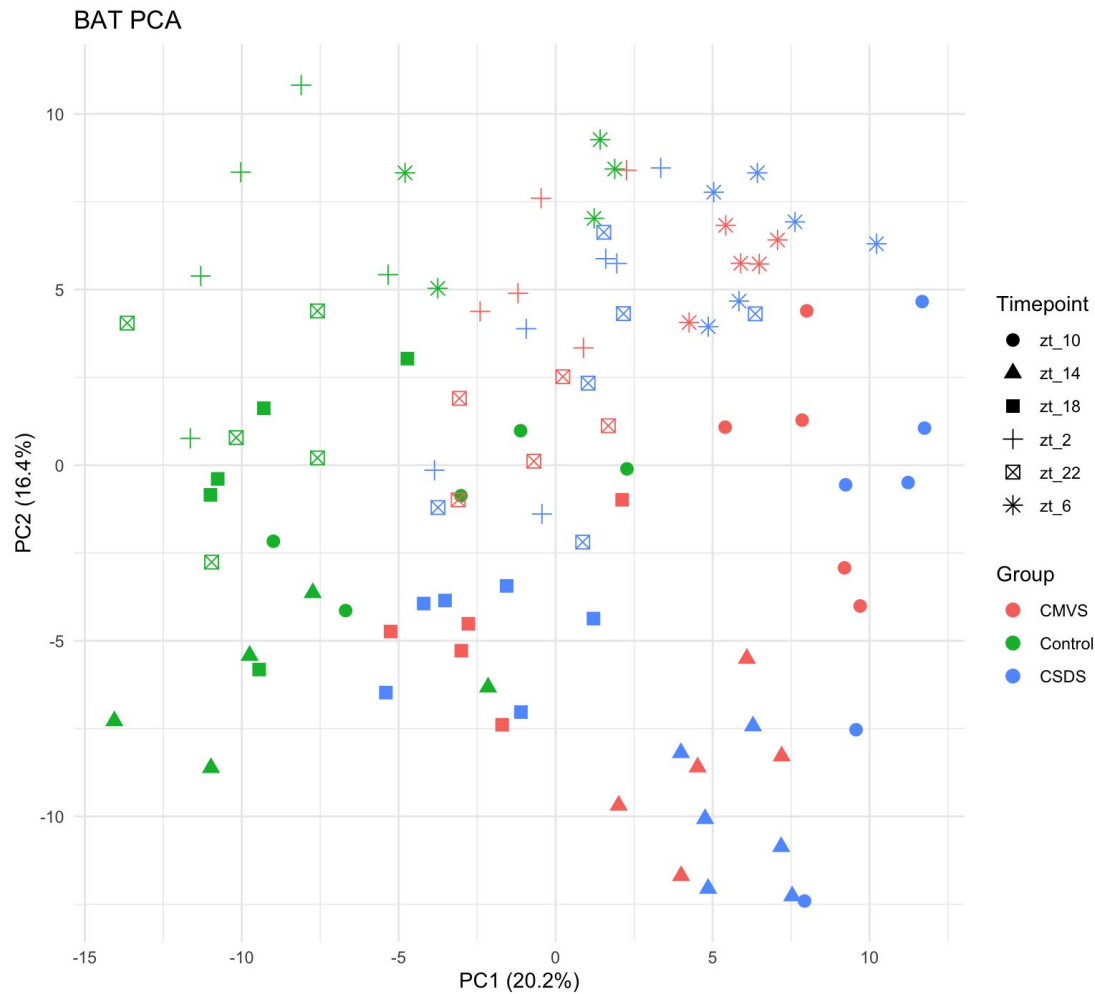
vW CSDS v Control



vW CSDS v CMVS

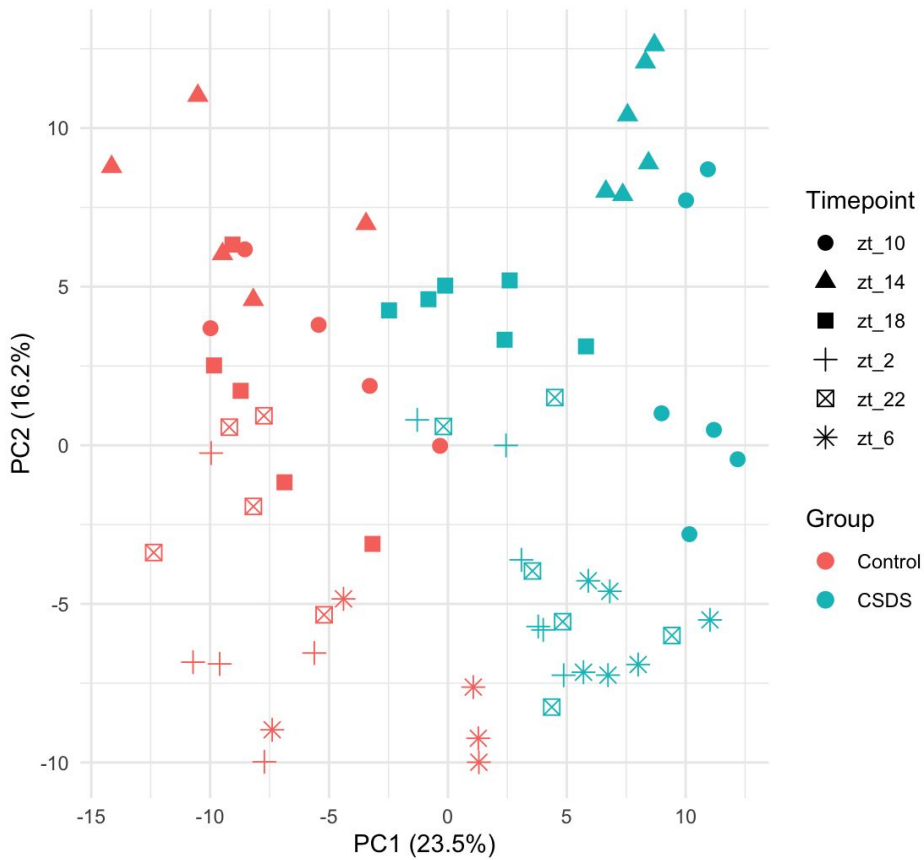


BAT PCA plot

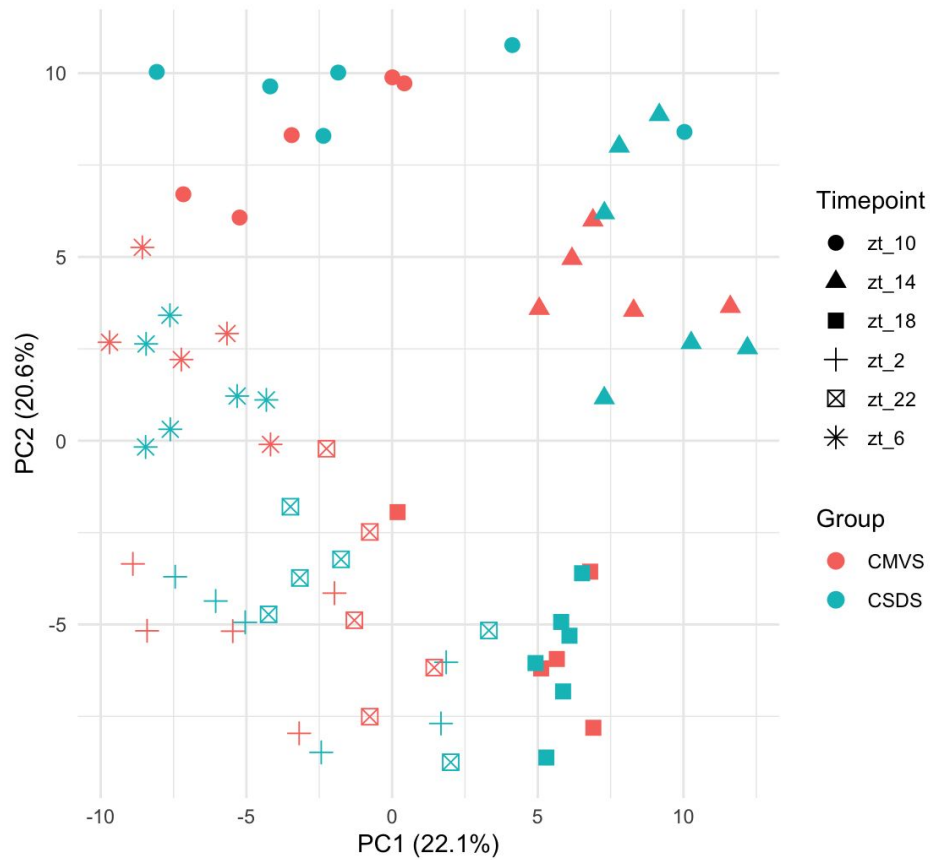


BAT PCA plots pairwise

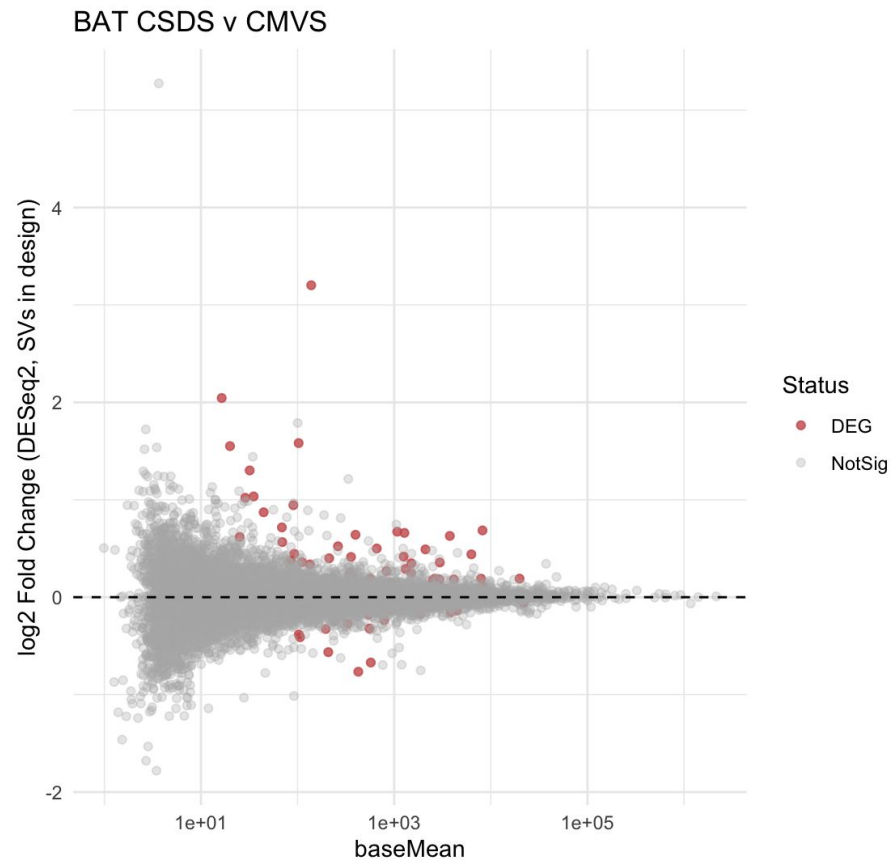
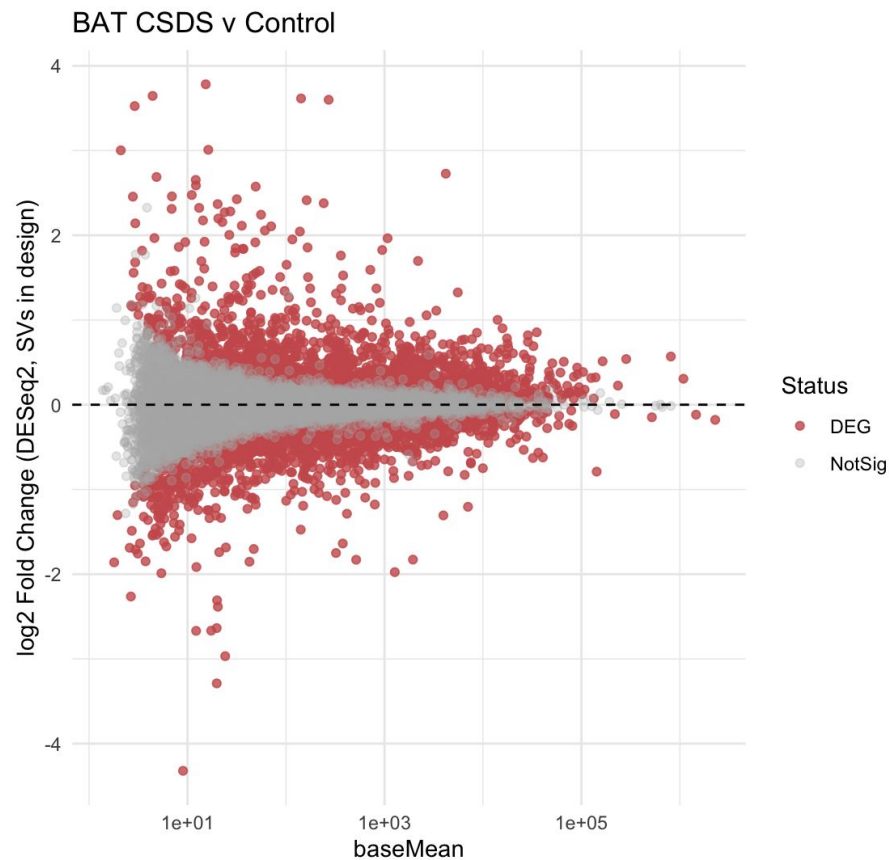
BAT CSDS v Control



BAT CSDS v CMVS

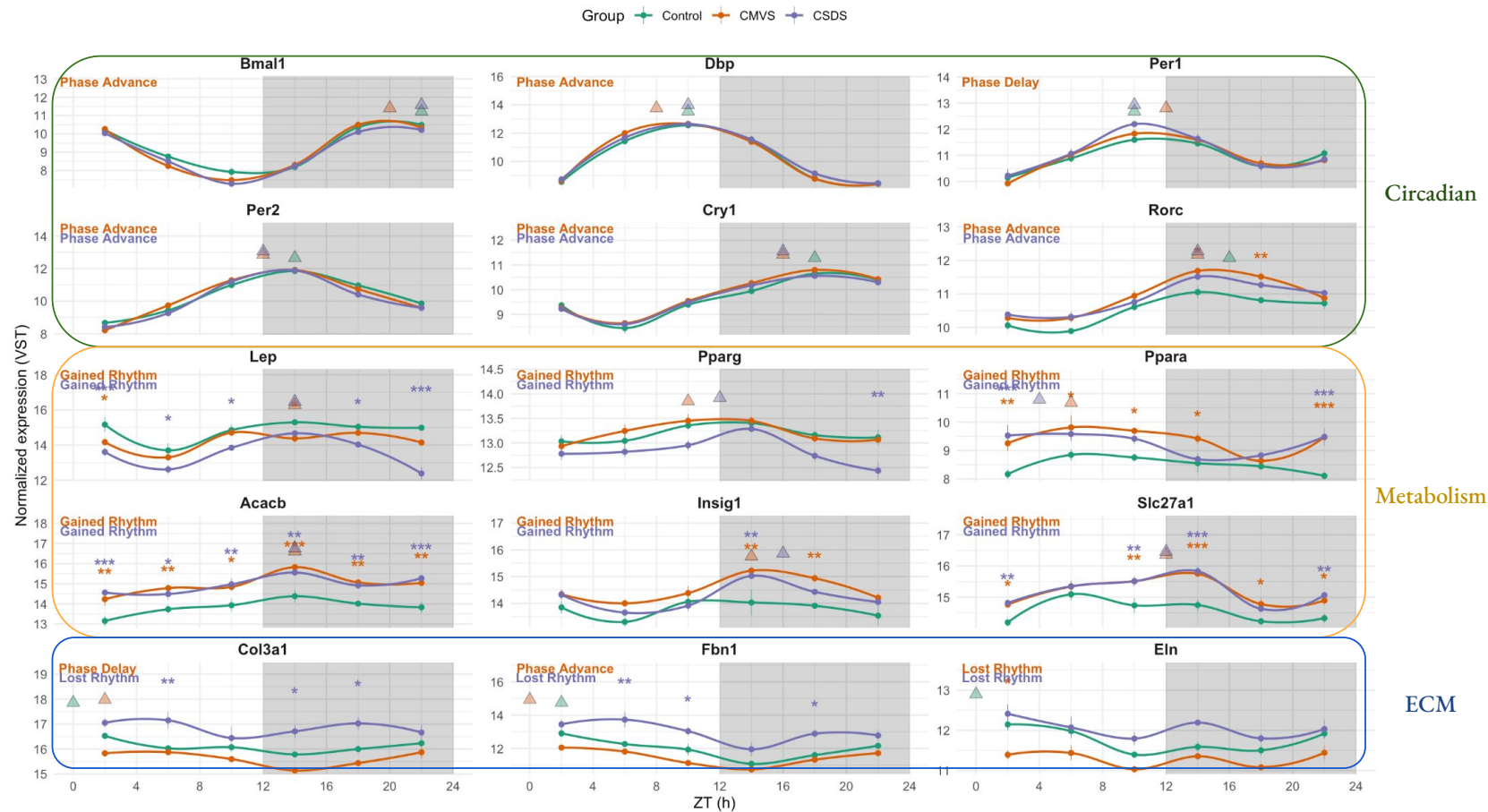


BAT MA plots



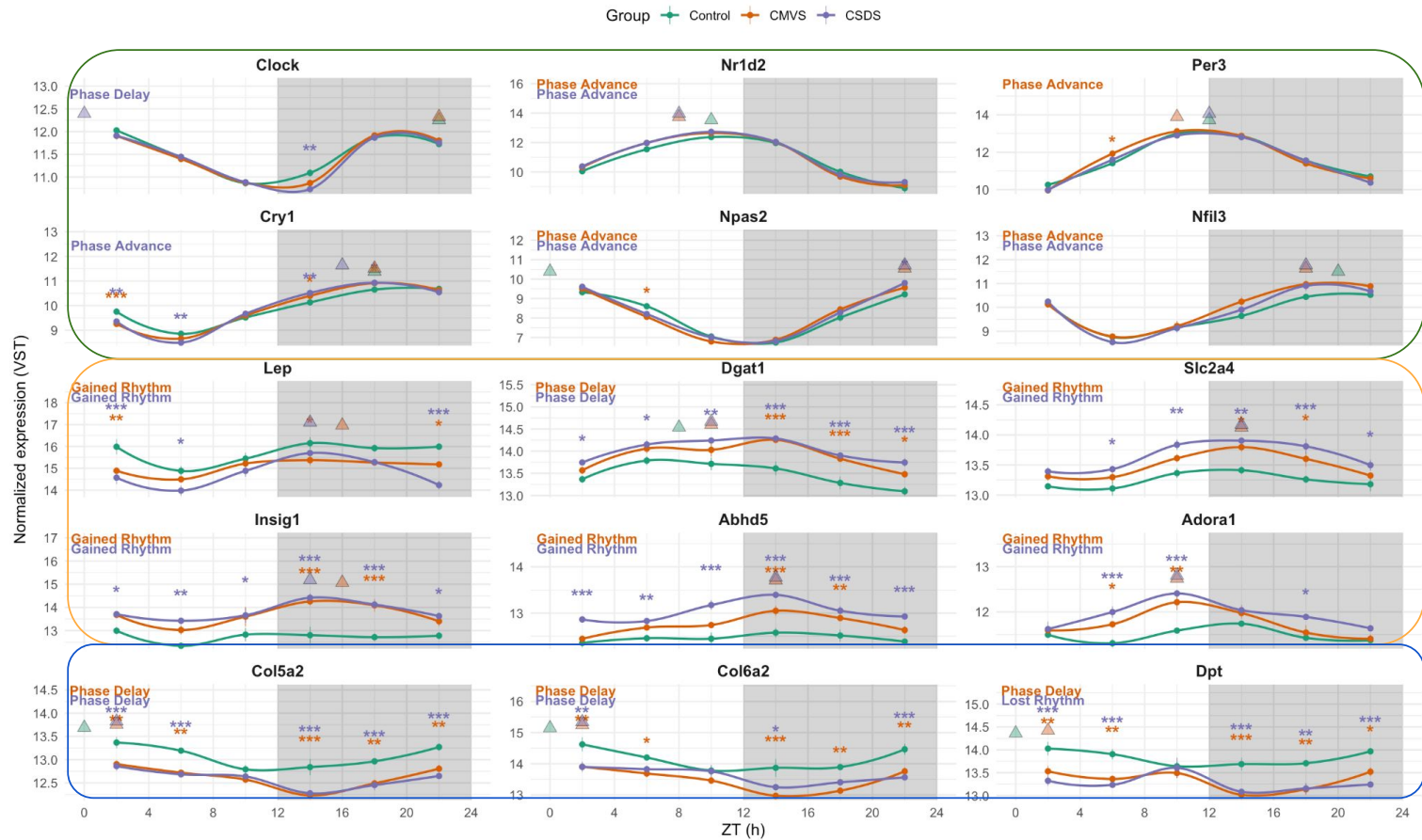
Stress affected clock, metabolism, and ECM genes in scW

scW genes — per-ZT post-hoc vs Control



Stress affected clock, metabolism, and ECM genes in vW

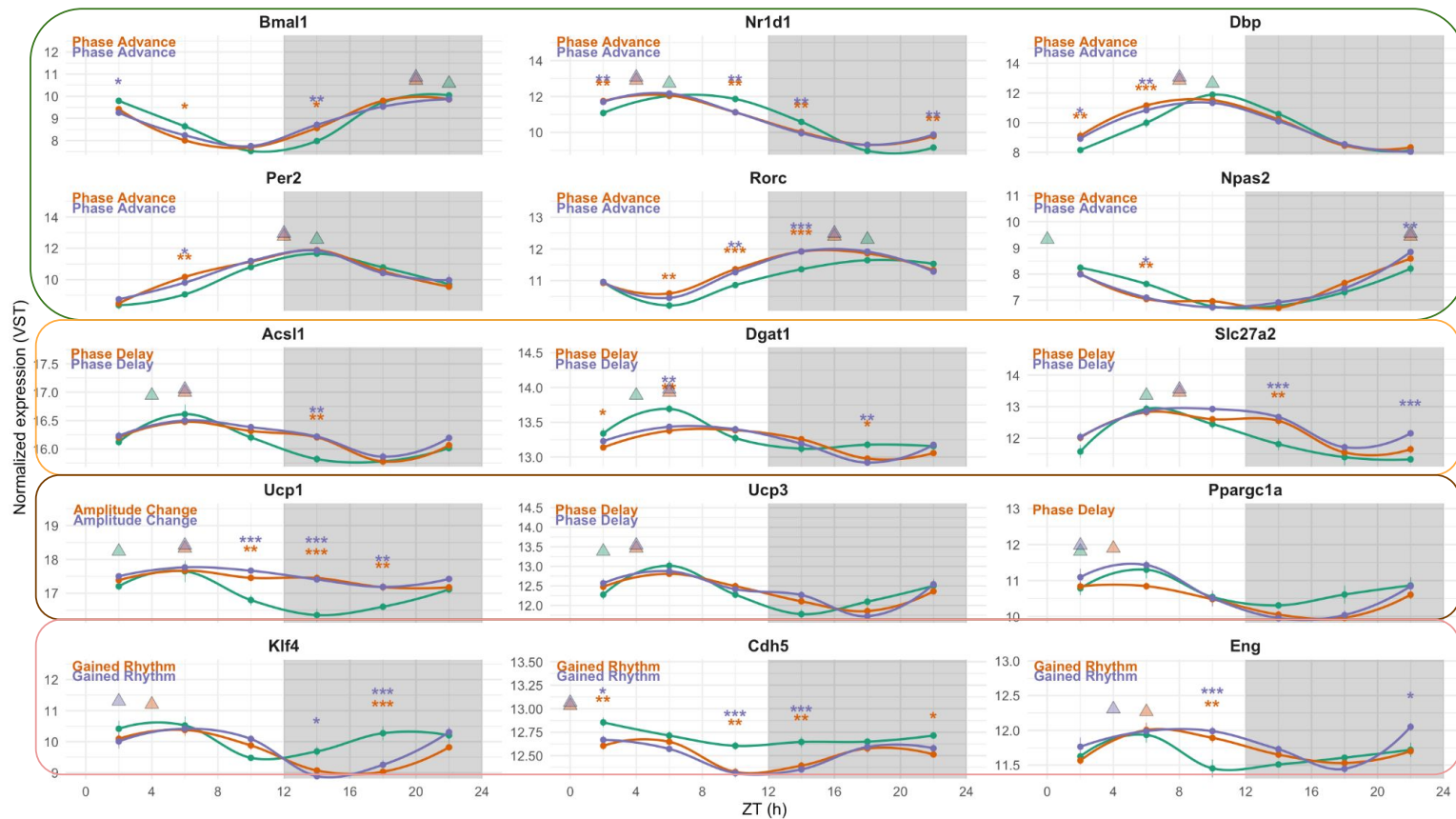
vW genes — per-ZT post-hoc vs Control



Stress affected clock, metabolism, thermogenesis, and angiogenesis genes in BAT

BAT genes — per-ZT post-hoc vs Control

Group ◆ Control ◆ CMVS ◆ CSDS



Circadian

Metabolism

Thermogenesis

Angiogenesis